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**Light-induced Phase Transitions
and NonLinear responses in
1D/2D Topological insulators**

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Introduction

The classification of phases of matter is one of the organizing themes of modern physics. For most of the twentieth century, this program rested on Landau's paradigm: phases are distinguished by the symmetries they break, and phase transitions are accompanied by the appearance or disappearance of a local order parameter. The discovery of the quantum Hall effect [1] and its theoretical explanation in terms of the topology of occupied Bloch bands [2] revealed that Landau's framework is fundamentally incomplete. Two insulators can share identical symmetries yet be genuinely distinct phases, separated by a boundary across which the bulk gap must close. The distinguishing quantity is not a local order parameter but a global, integer-valued topological invariant — the Chern number — computed from the geometric properties of electronic wavefunctions over the Brillouin zone.

This insight launched topological band theory. Zak identified the Berry phase as the natural topological invariant of one-dimensional insulators [3], while Berry established the geometric underpinnings of adiabatic quantum evolution [4]. The modern era began with the prediction [5, 6] and experimental realization [7, 8] of time-reversal-invariant topological insulators and has since expanded to topological semimetals, higher-order phases, and topological superconductors.

A parallel revolution has occurred in ultrafast laser physics. High Harmonic Generation (HHG) in solids encodes the Berry connection and curvature of occupied bands directly in the emitted harmonics [9, 10], making it an all-optical, femtosecond-timescale probe of band topology. The intersection of topological matter and ultrafast optics defines the subject of this thesis: Can a laser field actively modify topological invariants? What signatures do light-induced topological transitions leave in the nonlinear optical response? The theoretical answer is provided by Floquet theory — a trivial insulator can be driven into a topological phase by tuning the laser amplitude or frequency, yielding genuinely non-equilibrium *Floquet topological phase transitions* with no static counterpart.

Structure of this thesis. Chapter 1 establishes topological band theory: Berry connection, Berry curvature, Zak phase, winding number, Chern number, $\mathbb{Z}_2$ invariant, and the bulk–boundary correspondence. Chapter 2 develops the Floquet formalism: Peierls substitution, Floquet Hamiltonian, quasienergy spectrum in Sambe space, and the high-frequency effective Hamiltonian. Chapter 3 covers solid-state HHG: intraband and interband currents, the role of Berry geometry, and harmonic selection rules. Chapter 4 applies the framework to the SSH and Rice-Mele models, demonstrating that the even-to-odd harmonic ratio is an all-optical probe of the topological invariant with sharp features at Floquet transition points. Chapter 5 extends the analysis to the QWZ and Haldane models under circular driving, mapping Floquet Chern number phase diagrams and helicity-resolved HHG spectra. Three appendices provide supporting derivations. All numerical results were produced in Python; the code is available at <https://github.com/Hazem-74/Topological-Insulator-Models>.

Chapter 1: Topological Band Theory

1.1 Introduction

A central goal of modern condensed matter physics is to classify phases of matter not merely by their symmetries, but by their global topological properties. The discovery of topological insulators revealed that a bulk energy gap can coexist with robust, symmetry-protected boundary modes whose existence is guaranteed by the topology of the bulk electronic wavefunctions, rather than by any local property of the material. This chapter builds the theoretical language needed to understand and quantify this phenomenon.

We begin with Bloch's theorem and the standard framework of band theory, then introduce the geometric concepts of Berry phase and Berry curvature. These lead naturally to topological invariants — the Zak phase in one dimension, the Chern number in two dimensions, and an overview of the $\mathbb{Z}_2$ invariant for time-reversal-invariant systems. The chapter closes with the bulk–boundary correspondence, which connects bulk topology to the existence of protected edge states.

1.2 Bloch's Theorem and Crystal Momentum

A crystalline solid possesses discrete translational symmetry: the Hamiltonian H satisfies $[H, T_{\mathbf{a}}] = 0$, where $T_{\mathbf{a}}$ is the operator that translates by a lattice vector $\mathbf{a}$. Since $T_{\mathbf{a}}$ is unitary with eigenvalues of the form $e^{i\mathbf{k}\cdot\mathbf{a}}$, the energy eigenstates can be simultaneously labeled by a crystal momentum $\mathbf{k}$. This is the content of **Bloch's theorem**: in a periodic potential $V(\mathbf{r}) = V(\mathbf{r} + \mathbf{a})$, the energy eigenstates take the form

$$|\psi_{n\mathbf{k}}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}} |u_{n\mathbf{k}}\rangle, \quad (1.1)$$

where $|u_{n\mathbf{k}}\rangle$ is periodic over the unit cell, $u_{n\mathbf{k}}(\mathbf{r} + \mathbf{a}) = u_{n\mathbf{k}}(\mathbf{r})$, and n is the band index. The crystal momentum $\mathbf{k}$ plays the role of momentum in the lattice environment, but with an important distinction: it is defined only modulo a reciprocal lattice vector $\mathbf{G}$, so that $\mathbf{k} \equiv \mathbf{k} + \mathbf{G}$. Consequently, $\mathbf{k}$ is restricted to the first Brill-

loun zone (BZ), which in d dimensions has the topology of a d -dimensional torus T^d [11].

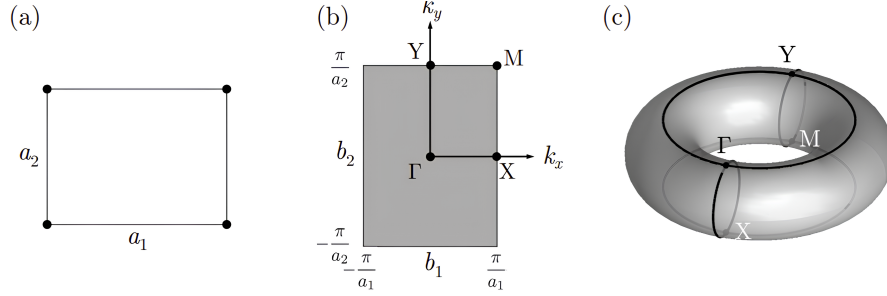


Figure 1.1: **(a)** Rectangular lattice with primitive vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. **(b)** Brillouin zone centered at $k_x = k_y = 0$. **(c)** Toroidal BZ geometry under periodic gauge, formed by rolling the rectangular BZ along the k_y -axis and joining its edges at $k_y = \pm\pi/a_2$ to form a torus.

Substituting (1.1) into the Schrödinger equation yields an eigenvalue problem for the cell-periodic part $|u_{n\mathbf{k}}\rangle$,

$$H(\mathbf{k}) |u_{n\mathbf{k}}\rangle = E_n(\mathbf{k}) |u_{n\mathbf{k}}\rangle, \quad H(r + a) = H(r). \quad (1.2)$$

where the Bloch Hamiltonian is defined as

$$H(\mathbf{k}) = e^{-i\mathbf{k}\cdot\mathbf{r}} H e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (1.3)$$

The band structure $E_n(\mathbf{k})$ — the dispersion of eigenvalues across the BZ — encodes nearly all static electronic properties of the solid. An insulator is characterized by a finite energy gap $E_G = \min_{\mathbf{k}}[E_{n+1}(\mathbf{k}) - E_n(\mathbf{k})] > 0$ separating the filled bands from the empty ones.

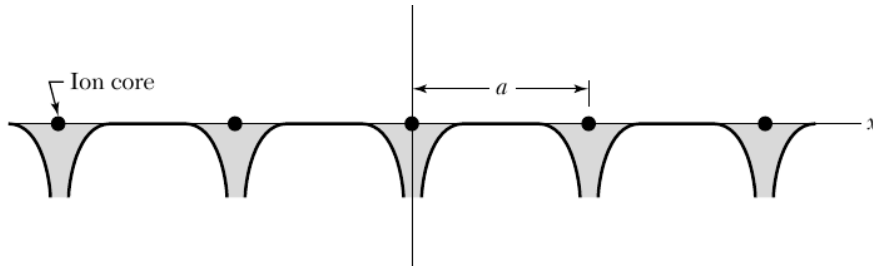


Figure 1.2: Periodic potential shows the electrons tighten on their sites.

1.3 The Tight-Binding Model

The tight-binding model offers the simplest concrete realization of band structure. The physical idea is that electrons are predominantly localized on atomic sites and

can tunnel to neighboring sites with an amplitude t , the hopping parameter. For a one-dimensional chain of N atoms with lattice spacing a and periodic boundary conditions, the Hamiltonian is

$$H = E_0 \sum_n |n\rangle \langle n| - t \sum_n (|n\rangle \langle n+1| + |n+1\rangle \langle n|), \quad (1.4)$$

where E_0 is the on-site energy and $|n\rangle$ denotes an electron localized at site n . The ansatz $\psi_n = e^{ikna}$ (a discrete Bloch wave) solves the Schrödinger equation with dispersion

$$E(k) = E_0 - 2t \cos(ka), \quad k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]. \quad (1.5)$$

This single band of width $4t$ illustrates several key features of band theory. For small k , the dispersion is approximately quadratic, $E(k) \approx \text{const} + ta^2 k^2$, recovering the free-particle form with an effective mass $m^* = \hbar^2/2ta^2$ that depends on the lattice rather than the bare electron mass. The periodicity of k enforces the Brillouin zone, and the N distinct k -values within it correspond exactly to the N orthogonal states of the Hilbert space [12].

Metals vs. Insulators

With multiple electrons, the Pauli exclusion principle requires each state to be occupied by at most one electron per spin. If the atom has valency Z , there are ZN electrons filling the N states from the bottom up. When a band is partially filled ($Z = 1$ in the single-band model), there are low-energy excitations available near the Fermi surface: the system is a **metal**. When a band is completely filled and a gap separates it from the next empty band ($Z = 2$ for a two-band model), no low-energy excitations exist and the system is a **band insulator**. This quantum-mechanical distinction — rooted in the band structure and the exclusion principle — is the foundation on which topological classification is built [12].

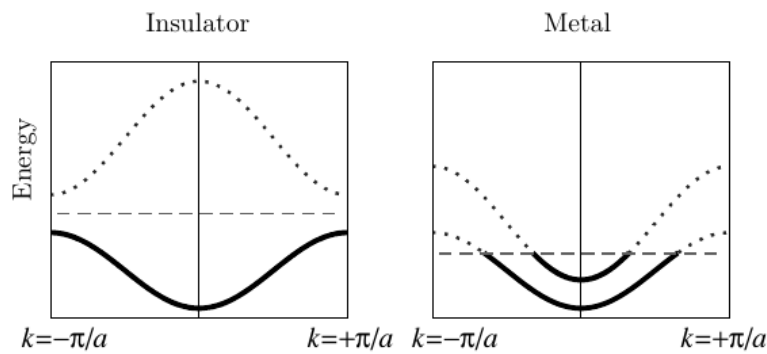


Figure 1.3: Band diagrams of metals vs insulators.

1.4 Geometric Phases

When a quantum system undergoes adiabatic evolution along a closed path C in parameter space (here, the Brillouin zone), the wavefunction acquires a geometric phase in addition to the dynamical phase. For a non-degenerate band n , this is known as the Berry phase.

1.4.1 Berry Connection and Berry Phase

The local geometric property of the Bloch state is described by the Berry connection (or Berry gauge potential), defined as:

$$\mathcal{A}_n(\mathbf{k}) = i\langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{n\mathbf{k}} \rangle. \quad (1.6)$$

The Berry connection is analogous to the vector potential in electromagnetism. Under a gauge transformation $|u_{n\mathbf{k}}\rangle \rightarrow e^{i\chi(\mathbf{k})}|u_{n\mathbf{k}}\rangle$, the connection transforms as $\mathcal{A}_n(\mathbf{k}) \rightarrow \mathcal{A}_n(\mathbf{k}) - \nabla_{\mathbf{k}}\chi(\mathbf{k})$.

The Berry phase γ_n acquired by the state when the crystal momentum $\mathbf{k}$ traverses a closed loop C in the Brillouin zone is the line integral of the Berry connection:

$$\gamma_n = \oint_C \mathcal{A}_n(\mathbf{k}) \cdot d\mathbf{k}. \quad (1.7)$$

While the Berry connection is gauge-dependent, the Berry phase modulo 2π is gauge-invariant and physically observable [13].

1.4.2 Berry Curvature and the Two-Band Model

By applying Stokes' theorem, the Berry phase can be expressed as the flux of the Berry curvature $\Omega_n(\mathbf{k})$ through a surface S bounded by C :

$$\gamma_n = \int_S \Omega_n(\mathbf{k}) \cdot d\mathbf{S}, \quad (1.8)$$

where the Berry curvature is defined as the curl of the Berry connection, $\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_n(\mathbf{k})$. In component form, using first-order perturbation theory, the Berry curvature can be written in a manifestly gauge-invariant form:

$$\Omega_{n,\mu\nu}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{\langle u_{n\mathbf{k}} | \partial_{k_\mu} H(\mathbf{k}) | u_{m\mathbf{k}} \rangle \langle u_{m\mathbf{k}} | \partial_{k_\nu} H(\mathbf{k}) | u_{n\mathbf{k}} \rangle}{(E_m(\mathbf{k}) - E_n(\mathbf{k}))^2}. \quad (1.9)$$

This expression reveals that the Berry curvature is large in regions of the Brillouin zone where bands are nearly degenerate, acting as a "magnetic field" in momentum

space [14].

It is highly instructive to evaluate the Berry phase for the simplest two-level Hamiltonian, which can be expressed in terms of Pauli matrices $\boldsymbol{\sigma}$ as:

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma} = \begin{pmatrix} d_z & d_x - id_y \\ d_x + id_y & -d_z \end{pmatrix}. \quad (1.10)$$

This Hamiltonian has eigenvalues $\pm|\mathbf{d}|$. We ignore terms proportional to the identity matrix, as they do not affect the eigenvectors, which depend only on the unit vector $\hat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|$. The vector $\hat{\mathbf{d}}$ can be viewed as a point on a Bloch sphere S^2 . A classic result derived by Berry is that for a loop C , the phase associated with the ground state is half the solid angle swept out by $\hat{\mathbf{d}}(\mathbf{k})$:

$$\gamma_C = \frac{1}{2}\Omega_{\text{solid}}. \quad (1.11)$$

The corresponding Berry curvature is given by the solid angle per unit area in $\mathbf{k}$ -space:

$$F = \frac{1}{2}\epsilon_{ij}\hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}). \quad (1.12)$$

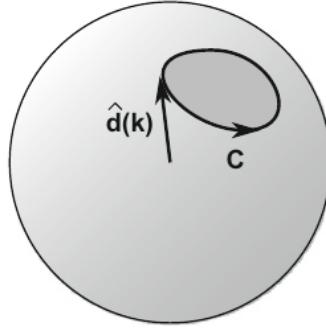


Figure 1.4: Geometric interpretation of the Berry phase in a two-band model as half the solid angle swept out by the $\hat{\mathbf{d}}(\mathbf{k})$ vector on the Bloch sphere.

1.5 Discrete Symmetries

To elucidate the topological invariance in materials, it is essential to discuss the discrete symmetries of the Hamiltonian and how they interplay with topological properties. The classification of topological phases relies heavily on the presence or absence of specific symmetries. We focus here on inversion, time-reversal, and chiral symmetries [15, 16].

1.5.1 Inversion (Parity) Symmetry

The inversion (or parity) operator $\mathcal{P}$ acts on a position eigenstate as $\mathcal{P}|\mathbf{r}\rangle = |-\mathbf{r}\rangle$. It follows that $\mathcal{P}^\dagger \mathbf{r} \mathcal{P} = -\mathbf{r}$ and $\mathcal{P}^\dagger \mathbf{p} \mathcal{P} = -\mathbf{p}$, meaning both position and momentum operators anticommute with $\mathcal{P}$. In momentum space, a Hamiltonian possessing inversion symmetry satisfies:

$$\mathcal{P}H(\mathbf{k})\mathcal{P}^{-1} = H(-\mathbf{k}). \quad (1.13)$$

As will be discussed in Section 1.6.3, the presence of inversion symmetry greatly simplifies the calculation of the $\mathbb{Z}_2$ topological invariant, reducing it to a product of parity eigenvalues at Time-Reversal Invariant Momenta (TRIM) points [15].

1.5.2 Time-Reversal Symmetry

Under time reversal, the time variable $t \rightarrow -t$, causing position $\mathbf{r}(t)$ to remain unchanged while momentum $\mathbf{p}(t) \rightarrow -\mathbf{p}(t)$ and angular momentum $\mathbf{L}(t) \rightarrow -\mathbf{L}(-t)$. In quantum mechanics, to preserve the Schrödinger equation $i\hbar\partial_t|\psi\rangle = H|\psi\rangle$ under $t \rightarrow -t$, the time-reversal operator $\mathcal{T}$ must be **anti-unitary**. It can be decomposed as $\mathcal{T} = U\mathcal{K}$, where U is a unitary operator and $\mathcal{K}$ is the complex conjugation operator.

For a spin-1/2 particle, the time-reversal operator takes the specific form $\mathcal{T} = -i\sigma_y\mathcal{K}$, which yields $\mathcal{T}^2 = -1$. This property leads directly to **Kramers' theorem**: for every eigenstate $|\psi\rangle$, the time-reversed state $\mathcal{T}|\psi\rangle$ is orthogonal to $|\psi\rangle$ and possesses the same energy, guaranteeing at least a two-fold degeneracy at TRIM points (points in the BZ where $\mathbf{k} \equiv -\mathbf{k} \pmod{\mathbf{G}}$) [15, 16].

For a bulk Hamiltonian in momentum space, time-reversal symmetry dictates:

$$H(\mathbf{k}) = UH(-\mathbf{k})^*U^\dagger, \quad (1.14)$$

which enforces the symmetry of the energy eigenvalues: $E_n(\mathbf{k}) = E_n(-\mathbf{k})$.

1.5.3 Chiral (Sublattice) Symmetry

Chiral symmetry, often referred to as sublattice symmetry in condensed matter physics, arises naturally in bipartite lattices (e.g., the SSH model or graphene with nearest-neighbor hopping), where sites of one sublattice only couple to sites of the other. Mathematically, a Hamiltonian possesses chiral symmetry if there exists a unitary operator Γ such that $\Gamma^2 = 1$ and:

$$\Gamma H(\mathbf{k})\Gamma^{-1} = -H(\mathbf{k}). \quad (1.15)$$

This symmetry forces the energy spectrum to be symmetric about zero energy ($E_n(\mathbf{k}) = -E_n(-\mathbf{k})$) and plays a critical role in quantizing the 1D Zak phase to 0 or π [14].

1.5.4 Particle-Hole Symmetry

Particle-hole symmetry (PHS) is an anti-unitary symmetry $\mathcal{C}$ that satisfies $\mathcal{C}H(\mathbf{k})\mathcal{C}^{-1} = -H(-\mathbf{k})$. While it is the defining symmetry of topological superconductors (described by the Bogoliubov–de Gennes Hamiltonian), it is not the primary focus of this thesis, which centers on normal-state topological insulators protected by TRS and chiral/inversion symmetries [16].

1.6 Topological Invariants

Topological invariants are global quantities derived from the Berry curvature that classify band structures into distinct topological phases. They are quantized, robust against continuous deformations of the Hamiltonian, and can only change when the bulk energy gap closes or a protecting symmetry is broken [17].

1.6.1 The Zak Phase (1D)

In one dimension, the natural closed loop in the BZ is the entire zone $k \in [-\pi/a, \pi/a]$, since the endpoints are identified by periodicity. The Berry phase acquired over this loop is the **Zak phase**:

$$\gamma_{\text{Zak}} = \int_{-\pi/a}^{\pi/a} A(k) dk = \int_{-\pi/a}^{\pi/a} (-i) \langle u_k | \partial_k | u_k \rangle dk. \quad (1.16)$$

Crucially, the Zak phase is quantized to 0 or π (modulo 2π) when the system possesses **chiral or inversion symmetry**. This quantization serves as the topological invariant distinguishing the two phases of 1D insulators, such as the Su-Schrieffer-Heeger (SSH) model. A Zak phase of π signals a topologically nontrivial phase [14].

1.6.2 The Chern Number (2D)

In two dimensions, the Berry curvature can be integrated over the entire BZ torus T^2 . The result is the **Chern number** (or TKNN invariant):

$$C_n = \frac{1}{2\pi} \int_{T^2} \Omega_n(\mathbf{k}) d^2\mathbf{k} \in \mathbb{Z}. \quad (1.17)$$

The quantization of $\mathcal{C}_n$ to an integer follows from the fact that the Berry curvature integrated over a closed surface must be a multiple of 2π , analogous to the Gauss–Bonnet theorem. Geometrically, referring to the two-band model in Eq. (1.10), the Chern number counts how many times the map $\mathbf{k} \mapsto \hat{\mathbf{d}}(\mathbf{k})$ wraps the BZ torus around the Bloch sphere.

A nonzero Chern number is the defining property of a Chern insulator (e.g., the Haldane model) and is directly related to the quantized Hall conductance via $\sigma_{xy} = \frac{e^2}{h} \sum_n \mathcal{C}_n$. Notably, a non-zero Chern number requires the **breaking of time-reversal symmetry**, as TRS forces $\Omega_n(\mathbf{k}) = -\Omega_n(-\mathbf{k})$, integrating to zero over the symmetric BZ [17].

1.6.3 The $\mathbb{Z}_2$ Invariant

In systems with time-reversal symmetry (TRS) and half-integer spin, $\mathcal{T}^2 = -1$. As established, TRS forces the Chern number of every band to vanish. However, a distinct $\mathbb{Z}_2$ topological invariant $\nu \in \{0, 1\}$ can still be defined, classifying **time-reversal-invariant topological insulators**.

When **inversion symmetry** is also present, the $\mathbb{Z}_2$ invariant takes a particularly simple form derived by Fu and Kane [18]:

$$(-1)^\nu = \prod_i \prod_{m=1}^N \xi_{2m}(\mathbf{\Lambda}_i), \quad (1.18)$$

where $\xi_{2m}(\mathbf{\Lambda}_i) = \pm 1$ are the parity eigenvalues of the $2m$ -th occupied Kramers pair at each of the TRIM points $\mathbf{\Lambda}_i$. A value $\nu = 1$ identifies a topological insulator protected by TRS, while $\nu = 0$ denotes the trivial phase [17].

1.7 Bulk–Boundary Correspondence

The topological invariants defined above are properties of the bulk Hamiltonian in momentum space. Yet their most striking physical consequence appears at real-space boundaries, governed by the **bulk–boundary correspondence**.

Consider an interface between two insulators with different bulk invariants. Since they are topologically distinct, there is no adiabatic path connecting their Hamiltonians that keeps the gap open everywhere. Therefore, as one moves across the interface in real space, the energy gap must close and reopen, necessarily producing localized gapless states at the boundary [14, 16].

More precisely:

- In a 1D system with a nontrivial Zak phase ($\gamma_{\text{Zak}} = \pi$), there exist zero-

energy states localized at the ends of an open chain. These are protected by chiral (sublattice) symmetry and cannot be removed unless the gap closes or the symmetry is broken.

- In a 2D Chern insulator with $\mathcal{C} \neq 0$, there are $|\mathcal{C}|$ chiral edge modes running along each boundary. These modes propagate in only one direction and are immune to backscattering, giving rise to the quantized Hall conductance.
- In a 2D $\mathbb{Z}_2$ topological insulator ($\nu = 1$), a single pair of helical edge modes exists at each boundary, related by time-reversal symmetry. Unlike the chiral modes of the Chern insulator, these are protected from backscattering only as long as TRS is preserved.

The bulk–boundary correspondence is one of the deepest organizing principles of topological phases of matter. In the subsequent chapters, we will examine how this interplay between bulk topology and boundary physics is manifested in dynamically driven systems. Specifically, we will explore how periodic light driving can induce effective Hamiltonians that dynamically break or restore these discrete symmetries, leading to light-induced topological phase transitions and distinct signatures in High Harmonic Generation (HHG).

Chapter 2: Floquet Theory and Light-Matter Coupling

2.1 Introduction

In Chapter 1, we established the framework for topological phases in static systems. However, modern experiments frequently subject materials to time-periodic external fields, such as laser irradiation. When the Hamiltonian becomes explicitly time-dependent, conventional band theory is insufficient. Floquet theory provides the mathematical framework for these periodically driven systems, playing a role analogous to Bloch theory for spatially periodic crystals [19, 20]. Just as Bloch's theorem reduces a spatial problem to a single unit cell, Floquet's theorem reduces a time-dependent problem to a single driving period, yielding an effective, time-independent Floquet Hamiltonian that captures the stroboscopic dynamics of the system.

This “Floquet engineering” allows one to modify the topological properties of a material without altering its chemical composition, enabling light-induced topological phase transitions and the creation of non-equilibrium phases [20, 21]. In this chapter, we introduce the Peierls substitution for light-matter coupling, state Floquet's theorem, and define the quasienergy spectrum within the extended Sambe space. We then generalize the concept of Berry phase to the Floquet representation, discuss the calculation of Floquet topological invariants, and conclude with the physical mechanisms that bridge this formalism to nonlinear optics.

2.2 Light-Matter Coupling in Lattice Systems

2.2.1 Minimal Coupling and the Peierls Substitution

To describe the interaction between electrons and an electromagnetic field in a solid, we start from the minimal coupling prescription in quantum mechanics. For a particle of charge $q = -e$ in an electromagnetic field described by the vector potential $\mathbf{A}(\mathbf{r}, t)$, the canonical momentum is shifted as $\hat{\mathbf{p}} \rightarrow \hat{\mathbf{p}} - q\mathbf{A}$ [11].

For a tight-binding Hamiltonian describing electrons hopping between lattice sites, the effect of a vector potential is captured by the **Peierls substitution** [20, 22]. Consider a generic tight-binding Hamiltonian in the absence of a field:

$$H_{\mathbf{A}=0} = \sum_{\mathbf{m}} \varepsilon_{\mathbf{m}} |\mathbf{m}\rangle \langle \mathbf{m}| + \sum_{\mathbf{m}, \mathbf{n}} t_{\mathbf{mn}} |\mathbf{m}\rangle \langle \mathbf{n}|. \quad (2.1)$$

In the presence of a vector potential $\mathbf{A}(\mathbf{r}, t)$, the hopping matrix elements acquire a complex phase factor:

$$t_{\mathbf{mn}} \rightarrow t_{\mathbf{mn}} \exp \left(-\frac{iq}{\hbar} \int_{\mathbf{r}_{\mathbf{n}}}^{\mathbf{r}_{\mathbf{m}}} d\mathbf{r} \cdot \mathbf{A}(\mathbf{r}, t) \right), \quad (2.2)$$

where the integral is taken along the straight-line path connecting sites $\mathbf{n}$ and $\mathbf{m}$. The detailed derivation of this substitution from the minimal coupling Hamiltonian is provided in Appendix B.1.

2.2.2 Time-Dependent Vector Potentials and Periodic Driving

In this thesis, we are primarily interested in time-periodic electromagnetic fields, such as those generated by laser irradiation. For a monochromatic field with frequency $\omega = 2\pi/T$, the vector potential takes the form:

$$\mathbf{A}(t) = \mathbf{A}_0 \cos(\omega t + \varphi), \quad (2.3)$$

where $\mathbf{A}_0$ is the amplitude and φ is the phase. For a spatially uniform driving field (valid when the sample size is much smaller than the wavelength of light), the Peierls substitution simplifies considerably. The hopping between nearest-neighbor sites separated by $\boldsymbol{\delta}$ becomes:

$$t \rightarrow t \exp \left(-\frac{iq}{\hbar} \mathbf{A}(t) \cdot \boldsymbol{\delta} \right). \quad (2.4)$$

This time-periodic modulation of the hopping amplitudes is the starting point for Floquet engineering of topological phases [20, 21].

2.2.3 The Current Operator

A fundamental consequence of the minimal coupling prescription is that the vector potential $\mathbf{A}$ and the current density $\mathbf{J}$ are conjugate variables. The macroscopic current operator is rigorously defined as the negative derivative of the Hamiltonian

with respect to the vector potential [23]:

$$\hat{\mathbf{J}} = -\frac{\partial \hat{H}}{\partial \mathbf{A}}. \quad (2.5)$$

For the tight-binding lattice models studied in this thesis, applying this definition to a Peierls-modified hopping term (Eq. 2.4) between sites j and i separated by the bond vector $\boldsymbol{\delta}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ yields:

$$\hat{\mathbf{J}}_{ij} = i\frac{q}{\hbar}\boldsymbol{\delta}_{ij}t_{ij}\exp\left(-i\frac{q}{\hbar}\mathbf{A}\cdot\boldsymbol{\delta}_{ij}\right)|i\rangle\langle j| + \text{h.c.} \quad (2.6)$$

Physically, the derivative extracts the bond vector $\boldsymbol{\delta}_{ij}$, demonstrating that the current is proportional to the hopping rate multiplied by the distance the charge travels. This operator is essential for computing the time-dependent macroscopic currents and the nonlinear optical response, including High Harmonic Generation (HHG), which is discussed in Chapter 3.

2.3 Floquet Formalism

2.3.1 Floquet's Theorem and Quasienergies

Consider the time-dependent Schrödinger equation for a Hamiltonian that is periodic in time with period T :

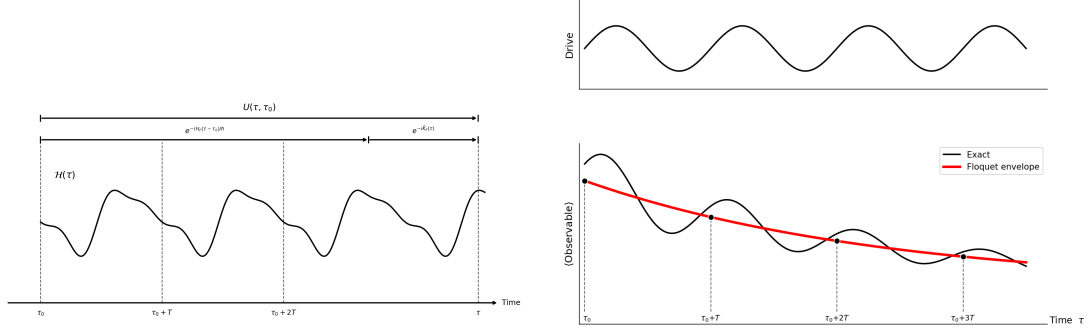
$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H(t)|\psi(t)\rangle, \quad H(t+T) = H(t). \quad (2.7)$$

The formal solution is governed by the time-evolution operator $U(t, t_0)$. Floquet's theorem states that this evolution operator can be decoupled into a slow, stroboscopic part and a fast, periodic part [19, 24]:

$$U(t, t_0) = e^{-i\hat{K}_F[t_0](t)}e^{-iH_F[t_0](t-t_0)/\hbar}, \quad (2.8)$$

where $H_F[t_0]$ is the time-independent **Floquet Hamiltonian** and $\hat{K}_F[t_0](t)$ is the periodic **kick operator** (or micromotion operator), satisfying $\hat{K}_F[t_0](t+T) = \hat{K}_F[t_0](t)$.

Equation (2.8) achieves the desired separation of time scales: the slow dynamics from one period to the next is governed by $H_F[t_0]$, while the fast “micromotion” within each period is captured by $\hat{K}_F[t_0](t)$ [19, 25]. Let $|n[t_0]\rangle$ be an eigenstate of the Floquet Hamiltonian with eigenvalue ε_n , called the **quasienergy**. Because the time-evolution operator over one period is $e^{-iH_FT/\hbar}$, the quasienergies are only



(a) Floquet's theorem illustrated. The slow dynamics (from one period T to the next) is captured by the Floquet Hamiltonian H_F , whereas the fast “micromotion” within a period is governed by the fast motion operator $e^{-i\hat{K}_F(\tau)}$.

(b) Time-evolution of an observable under periodic driving. The red line shows the slow dynamics governed by H_F , while the black line shows the exact dynamics including fast micromotion, which vanishes at stroboscopic times.

defined modulo the driving quantum $\hbar\omega$:

$$\varepsilon_n \equiv \varepsilon_n + m\hbar\omega, \quad m \in \mathbb{Z}. \quad (2.9)$$

We therefore restrict quasienergies to the **Floquet-Brillouin zone** $\varepsilon \in [-\hbar\omega/2, \hbar\omega/2)$.

The corresponding **Floquet states** are $|\psi_n[t_0](t)\rangle = e^{-i\varepsilon_n(t-t_0)/\hbar} |u_n[t_0](t)\rangle$, where $|u_n[t_0](t)\rangle \equiv e^{-i\hat{K}_F[t_0](t)} |n[t_0]\rangle$ are the periodic Floquet modes [20, 24].

2.3.2 The Sambe Space and Floquet-Bloch States

When a crystalline system is subjected to a spatially uniform, time-periodic driving field, the Hamiltonian possesses both discrete spatial translation symmetry and discrete time translation symmetry. This leads to the **Floquet-Bloch ansatz** [26, 27]:

$$|\Psi_{\alpha,\mathbf{k}}(\mathbf{r}, t)\rangle = e^{i\mathbf{k}\cdot\mathbf{r} - i\varepsilon_{\alpha,\mathbf{k}}t/\hbar} |u_{\alpha,\mathbf{k}}(\mathbf{r}, t)\rangle, \quad (2.10)$$

where $|u_{\alpha,\mathbf{k}}(\mathbf{r}, t)\rangle$ is periodic in both real space and time.

To formalize this mathematically, we introduce the **Sambe space** (or extended Hilbert space) $\mathcal{S} = \mathcal{H} \otimes \mathcal{T}$, where $\mathcal{T}$ is the space of T -periodic functions of time [24]. In this extended space, the time-dependent Schrödinger equation is elegantly mapped to a time-independent eigenvalue problem:

$$\mathcal{H}(\mathbf{k}, t) |u_{\alpha,\mathbf{k}}\rangle = \varepsilon_{\alpha,\mathbf{k}} |u_{\alpha,\mathbf{k}}\rangle, \quad (2.11)$$

where the **Floquet operator** is defined as $\mathcal{H}(\mathbf{k}, t) = H(\mathbf{k}, t) - i\hbar \frac{\partial}{\partial t}$. The inner product in Sambe space is defined by averaging over one driving period: $\langle\langle u|v\rangle\rangle = \frac{1}{T} \int_0^T dt \langle u(t)|v(t)\rangle$.

2.3.3 The High-Frequency Expansion

In most practical cases, it is impossible to find the exact Floquet Hamiltonian H_F or effective Hamiltonian H_{eff} in closed form from the infinite-dimensional Sambe matrix. However, when the driving frequency ω is much larger than all other energy scales in the system, we can employ a systematic **high-frequency expansion** (HFE) in inverse powers of ω [25, 28].

We begin by expanding the time-periodic Hamiltonian in Fourier components: $H(t) = \sum_{m=-\infty}^{\infty} H_m e^{im\omega t}$. The first few terms of the high-frequency expansion for the effective Hamiltonian are:

$$H_{\text{eff}}^{(0)} = H_0 = \frac{1}{T} \int_0^T dt H(t), \quad (2.12)$$

$$H_{\text{eff}}^{(1)} = \frac{1}{\hbar\omega} \sum_{m=1}^{\infty} \frac{1}{m} [H_m, H_{-m}]. \quad (2.13)$$

The zeroth-order term is simply the time-averaged Hamiltonian. The first-order correction involves commutators of Fourier components and captures the leading effects of the periodic driving, such as the generation of complex hopping amplitudes that break time-reversal symmetry. This expansion provides the practical toolset for calculating the effective Hamiltonians whose topological invariants we will compute in Section 2.5.

2.4 Geometric Phases in Floquet Systems

In Chapter 1, we established the geometric phase and Berry curvature for static band structures. When the system is subjected to a time-periodic drive, the electronic states are described by Floquet modes rather than static Bloch states. It is therefore essential to generalize the concept of Berry phase to the Floquet representation [29, 30].

2.4.1 Floquet Berry Connection and Berry Phase

For a periodically driven lattice, the Floquet modes $|u_n(\mathbf{k}, t)\rangle$ are periodic in time and depend parametrically on the crystal momentum $\mathbf{k}$. To define a geometric connection, we must utilize the inner product appropriate for the extended Hilbert space (Sambe space):

$$\langle\langle u_m(\mathbf{k}) | u_n(\mathbf{k}) \rangle\rangle = \frac{1}{T} \int_0^T dt \langle u_m(\mathbf{k}, t) | u_n(\mathbf{k}, t) \rangle = \delta_{mn}. \quad (2.14)$$

The **Floquet Berry connection** for the n -th Floquet band is then defined analogously to the static case, but using the Sambe inner product [29, 30]:

$$\mathcal{A}_n(\mathbf{k}) = i \langle \langle u_n(\mathbf{k}) | \nabla_{\mathbf{k}} | u_n(\mathbf{k}) \rangle \rangle = \frac{i}{T} \int_0^T dt \langle u_n(\mathbf{k}, t) | \nabla_{\mathbf{k}} | u_n(\mathbf{k}, t) \rangle. \quad (2.15)$$

This expression reveals that the Floquet Berry connection is the time-averaged Berry connection of the instantaneous Floquet modes. The **Floquet Berry phase** acquired by the n -th band when the crystal momentum traverses a closed loop C in the Brillouin zone is the line integral of this connection: $\gamma_n^F = \oint_C \mathcal{A}_n(\mathbf{k}) \cdot d\mathbf{k}$.

2.4.2 Floquet Berry Curvature and the Two-Band Model

The **Floquet Berry curvature** is defined as the curl of the Floquet Berry connection: $\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_n(\mathbf{k})$. Using the completeness of the Floquet modes in the Sambe space, this can be expressed in a manifestly gauge-invariant form:

$$\Omega_{n,\mu\nu}(\mathbf{k}) = -2 \operatorname{Im} \sum_{m \neq n} \frac{\langle \langle u_n | \partial_{k_\mu} \mathcal{H} | u_m \rangle \rangle \langle \langle u_m | \partial_{k_\nu} \mathcal{H} | u_n \rangle \rangle}{(\varepsilon_m - \varepsilon_n)^2}, \quad (2.16)$$

where $\mathcal{H}$ is the Floquet operator in the Sambe space, and ε_n are the quasienergies.

For a driven two-band system, the effective Floquet Hamiltonian can be parametrized by a time-independent vector $\mathbf{d}_F(\mathbf{k})$: $H_F(\mathbf{k}) = \mathbf{d}_F(\mathbf{k}) \cdot \boldsymbol{\sigma}$. Following the exact same geometric derivation as in the static case, the Floquet Berry curvature is given by half the solid angle swept out by the unit vector $\hat{\mathbf{d}}_F(\mathbf{k}) = \mathbf{d}_F(\mathbf{k}) / |\mathbf{d}_F(\mathbf{k})|$ on the Bloch sphere:

$$F_F = \frac{1}{2} \epsilon_{ij} \hat{\mathbf{d}}_F \cdot \left(\partial_{k_i} \hat{\mathbf{d}}_F \times \partial_{k_j} \hat{\mathbf{d}}_F \right). \quad (2.17)$$

This profound result demonstrates that the complex, time-dependent dynamics of a driven two-band system can be entirely mapped onto a static geometric problem.

2.5 Floquet Topological Invariants

2.5.1 Effective Hamiltonian and Topological Classification

One of the most powerful applications of Floquet theory is the engineering of topological phases via periodic driving. The effective Floquet Hamiltonian H_{eff} (derived via the high-frequency expansion in Section 2.3.3) is a time-independent operator that can be analyzed using the same topological classification schemes developed for equilibrium systems [20, 21, 31].

Consider a two-band insulator described by a static Bloch Hamiltonian $H_0(\mathbf{k}) =$

$\varepsilon_0(\mathbf{k})\mathbb{K}_{2\times 2} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$. Under periodic driving, the effective Floquet Hamiltonian can be parametrized similarly:

$$H_{\text{eff}}(\mathbf{k}) = \varepsilon_F(\mathbf{k})\mathbb{K}_{2\times 2} + \mathbf{n}(\mathbf{k}) \cdot \boldsymbol{\sigma}. \quad (2.18)$$

Provided that $|\mathbf{n}(\mathbf{k})| \neq 0$ throughout the Brillouin zone (i.e., the Floquet spectrum is gapped), we can define the unit vector $\hat{\mathbf{n}}(\mathbf{k}) = \mathbf{n}(\mathbf{k})/|\mathbf{n}(\mathbf{k})|$. The winding of this vector defines the topological invariants of the driven system.

2.5.2 The Floquet Winding Number (1D)

For one-dimensional systems with chiral (sublattice) symmetry, the Hamiltonian takes the off-diagonal form $H(k) = \begin{pmatrix} 0 & q(k) \\ q^\dagger(k) & 0 \end{pmatrix}$. For the effective Floquet Hamiltonian with chiral symmetry, we can similarly write $H_{\text{eff}}(k)$ with a complex function $q_F(k)$. The **Floquet winding number** is then defined as [14, 31]:

$$W_F = \frac{1}{2\pi i} \int_{-\pi/a}^{\pi/a} dk \frac{d}{dk} \ln \det q_F(k) = \frac{1}{2\pi} \int_{-\pi/a}^{\pi/a} dk \frac{d\phi_F(k)}{dk}, \quad (2.19)$$

where $\phi_F(k)$ is the phase of $\det q_F(k)$. This integer counts how many times the vector $\hat{\mathbf{n}}(k)$ winds around the origin in the (n_x, n_y) plane as k traverses the 1D Brillouin zone. For driven systems like the Su-Schrieffer-Heeger (SSH) model, W_F can differ from the static winding number W_0 ($\sim$ Zak phase), demonstrating that periodic driving can induce topological phase transitions in 1D.

2.5.3 The Floquet Chern Number (2D)

For two-dimensional systems, the winding of $\hat{\mathbf{n}}(\mathbf{k})$ over the Brillouin zone torus T^2 defines the **Floquet Chern number** [20]:

$$C_F = \frac{1}{4\pi} \int_{\text{BZ}} d^2\mathbf{k} \hat{\mathbf{n}}(\mathbf{k}) \cdot \left(\frac{\partial \hat{\mathbf{n}}(\mathbf{k})}{\partial k_x} \times \frac{\partial \hat{\mathbf{n}}(\mathbf{k})}{\partial k_y} \right). \quad (2.20)$$

This is formally identical to the Chern number for static systems, but computed from the Floquet Hamiltonian. Remarkably, $C_F \neq 0$ is possible even when the static Chern number $C_0 = 0$, meaning that periodic driving can induce a topological phase transition from a trivial insulator to a Chern insulator [20, 21, 32].

2.5.4 Floquet $\mathbb{Z}_2$ Invariants

When the driven system preserves time-reversal symmetry (TRS) with $\mathcal{T}^2 = -1$, the Floquet Chern numbers must vanish. However, a $\mathbb{Z}_2$ topological invariant can still be defined [20, 33]. If the system also possesses inversion symmetry, the Floquet $\mathbb{Z}_2$ invariant ν_F can be computed efficiently from the parity eigenvalues of the Floquet bands at the time-reversal invariant momenta (TRIM) [18, 20]:

$$(-1)^{\nu_F} = \prod_i \prod_{m=1}^N \xi_{2m}(\mathbf{\Lambda}_i), \quad (2.21)$$

where $\xi_{2m}(\mathbf{\Lambda}_i) = \pm 1$ are the parity eigenvalues of the occupied Floquet-Kramers pairs at each TRIM point $\mathbf{\Lambda}_i$. A non-trivial invariant ($\nu_F = 1$) indicates the presence of helical edge states protected by time-reversal symmetry.

2.6 Physical Mechanisms and the Bridge to Non-linear Optics

The ability to alter the topological properties of a material via light relies on a few fundamental physical mechanisms, which are mathematically captured by the high-frequency expansion and the micromotion operator.

2.6.1 Dynamical Symmetry Breaking and Gap Renormalization

A static Hamiltonian often possesses discrete symmetries that protect its topological classification. A periodically driven system, however, allows for the *dynamical breaking* of these symmetries. For instance, circularly polarized light explicitly breaks time-reversal symmetry by introducing a preferred direction of rotation. Mathematically, this is captured by the first-order term in the high-frequency expansion (Eq. (2.13)), where the commutator $[H_m, H_{-m}]$ generates complex, direction-dependent hopping amplitudes, driving a trivial insulator into a topological insulator [20, 21].

Furthermore, when the driving frequency is near-resonant with the intrinsic band gap ($\hbar\omega \approx E_g$), the electromagnetic field strongly hybridizes the valence and conduction bands. This interaction manifests as an avoided crossing in the quasienergy spectrum, effectively renormalizing the band gap (the solid-state analog of the AC Stark shift) [32]. In topological systems, this light-induced gap renormalization can force the gap to close and subsequently reopen with an in-

verted band ordering, signaling a Floquet topological phase transition [34].

2.6.2 Micromotion and the Bridge to Nonlinear Optics

It is crucial to distinguish between the stroboscopic dynamics governed by H_{eff} and the continuous time evolution. While H_{eff} captures the long-term, period-averaged topology, the system simultaneously exhibits fast, periodic oscillations within each driving cycle, governed by the kick operator $\hat{K}(t)$. This fast dynamics is known as **micromotion** [25].

Although micromotion averages to zero over a full period, it is the primary driver of highly nonlinear interband and intraband electron dynamics. As we will explore in Chapter 3, it is precisely this ultrafast micromotion—driven by the strong electric field of the laser—that underpins the rich optical signatures and topological fingerprints observed in high-harmonic generation (HHG). The Floquet formalism thus provides not only the topological classification of the driven bands but also the fundamental dynamical equations required to calculate the nonlinear optical response.

Chapter 3: Nonlinear Dynamics and High Harmonic Generation

3.1 Introduction

In the preceding chapters, we established the topological classification of static band structures (Chapter 1) and developed the formalism for understanding how periodic light driving modifies these topological phases (Chapter 2). However, to experimentally probe and utilize these light-induced topological states, we must understand how the material responds optically to intense electromagnetic fields.

When the electric field of light becomes comparable to the internal atomic fields of a material, the optical response ceases to be linear. This regime of **non-linear optics** gives rise to a wealth of phenomena, the most dramatic of which is High Harmonic Generation (HHG). While HHG was originally discovered and extensively studied in atomic gases, its observation in bulk crystals in 2011 launched a new field of attosecond condensed matter physics. In solids, the periodic lattice and the geometric properties of the band structure fundamentally alter the HHG process, making it a powerful, all-optical probe of topological invariants and band geometry.

In this chapter, we first review the fundamentals of nonlinear optics and the physical origin of optical nonlinearities. We then transition to the non-perturbative regime of HHG, contrasting the gas-phase three-step model with the solid-state paradigm. Finally, we decompose the solid-state HHG current into interband and intraband contributions, explicitly demonstrating how the Berry connection and Berry curvature govern the nonlinear electron dynamics and leave distinct topological fingerprints in the emitted harmonic spectrum.

3.2 Fundamentals of Nonlinear Optics

3.2.1 The Nonlinear Susceptibility Tensor

In linear optics, the macroscopic polarization $\mathbf{P}(t)$ induced in a medium by an applied electric field $\mathbf{E}(t)$ is assumed to be directly proportional to the field strength: $\mathbf{P}(t) = \epsilon_0 \chi^{(1)} \mathbf{E}(t)$, where $\chi^{(1)}$ is the linear susceptibility. However, when the applied field is sufficiently intense, this linear approximation breaks down. The polarization must then be expressed as a power series expansion in the electric field [35, 36]:

$$\mathbf{P}(t) = \epsilon_0 \left(\chi^{(1)} \mathbf{E}(t) + \chi^{(2)} \mathbf{E}^2(t) + \chi^{(3)} \mathbf{E}^3(t) + \cdots \right), \quad (3.1)$$

where $\chi^{(n)}$ is the n -th order nonlinear susceptibility tensor of rank $n + 1$. For example, the second-order polarization is written in component form as $P_i^{(2)}(t) = \epsilon_0 \sum_{j,k} \chi_{ijk}^{(2)} E_j(t) E_k(t)$. The presence of these higher-order terms signifies that the medium's response is no longer instantaneous or purely harmonic, leading to the generation of new frequency components.

While this perturbative expansion successfully describes standard nonlinear optical processes, it loses its validity when the field strength approaches the characteristic atomic binding field ($E_{\text{at}} \approx 5.14 \times 10^{11}$ V/m). In this extreme regime, the perturbative expansion diverges, and the system enters the non-perturbative domain characterized by High Harmonic Generation (HHG) [35, 37].

3.2.2 Physical Origin of Optical Nonlinearities

The microscopic origin of these nonlinearities can be understood classically through the anharmonic oscillator model. An electron bound to an atom is held in a potential well $U(x)$. For weak fields, the displacement x is small, and the potential is well-approximated by a harmonic (quadratic) well, yielding a linear restoring force. However, for intense laser fields, the electron is driven far from equilibrium, and the anharmonicity of the true atomic potential becomes significant [36, 38]:

$$U(x) = \frac{1}{2} m \omega_0^2 x^2 + \frac{1}{3} m C_3 x^3 + \frac{1}{4} m C_4 x^4 + \cdots, \quad (3.2)$$

where ω_0 is the resonant frequency, m is the electron mass, and C_3, C_4 are anharmonic coefficients. The resulting restoring force $F = -dU/dx$ is no longer strictly proportional to the displacement. When driven by an oscillating electric field, this anharmonic motion produces a polarization that contains higher-order harmonics of the driving frequency, directly corresponding to the $\chi^{(2)}$ and $\chi^{(3)}$ terms in Eq.

(3.1). A detailed classical derivation of this mechanism, leading to Miller's Rule for the second-order susceptibility, is provided in Appendix C.1.

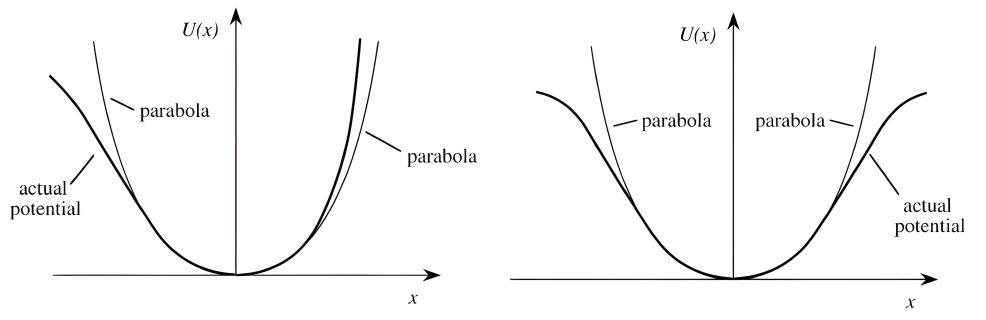


Figure 3.1: Potential energy functions for the anharmonic oscillator model.

3.2.3 Standard Nonlinear Optical Processes

The power-series expansion of the polarization naturally describes a variety of frequency-mixing processes. If the applied field consists of multiple frequencies, the nonlinear terms generate new frequencies corresponding to the sums and differences of the input fields [35]:

- **Second-Harmonic Generation (SHG):** Arising from $\chi^{(2)}$, two photons of frequency ω are annihilated to create one photon at 2ω . Crucially, in the dipole approximation, $\chi^{(2)}$ vanishes identically in any medium possessing inversion symmetry (centrosymmetric media).
- **Sum- and Difference-Frequency Generation (SFG/DFG):** Also governed by $\chi^{(2)}$, these processes mix two distinct input frequencies ω_1 and ω_2 to generate radiation at $\omega_1 + \omega_2$ or $\omega_1 - \omega_2$.
- **Third-Harmonic Generation (THG) and Four-Wave Mixing:** Governed by $\chi^{(3)}$, these processes can occur in all media, including gases, liquids, and centrosymmetric crystals, because the third-rank tensor does not vanish under inversion symmetry.

While these perturbative nonlinear processes are foundational, they describe regimes where the laser field is still a weak perturbation to the atomic potential. As the field strength approaches the atomic binding limit, the power-series expansion fails, necessitating the non-perturbative framework of High Harmonic Generation discussed in the next section.

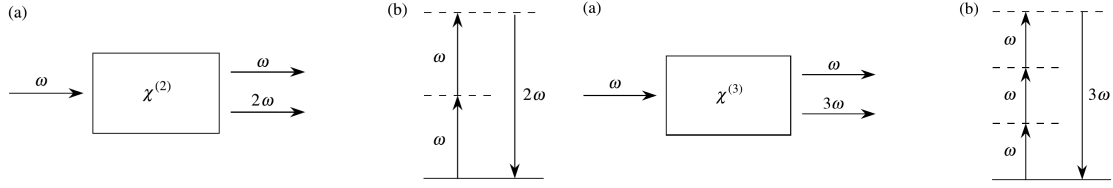


Figure 3.2: Second & Third-harmonic generation. (a) Geometry of the interaction. (b) Energy-level description.

3.3 High Harmonic Generation: From Gases to Solids

3.3.1 The Gas-Phase Paradigm

High Harmonic Generation was first observed in atomic gases in the late 1980s [37, 39]. The underlying physics is elegantly captured by the semiclassical **three-step model** proposed by Corkum [40]:

1. **Tunnel Ionization:** The intense laser field suppresses the Coulomb barrier, allowing an electron to tunnel into the continuum.
2. **Acceleration (Propagation):** The freed electron is accelerated away from the parent ion by the oscillating laser field, gaining kinetic energy (ponderomotive energy $U_p \propto I\lambda^2$).
3. **Recollision:** When the field reverses, the electron is driven back to recombine with its parent ion, emitting a high-energy photon with energy $\hbar\omega = I_p + K_{\text{kin}}$, where I_p is the ionization potential and K_{kin} is the kinetic energy upon return.

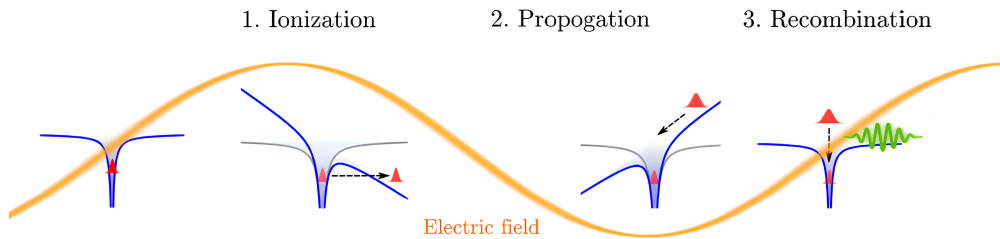


Figure 3.3: Schematic diagram of the three-step model for high-order harmonic generation in gases.

This model predicts a maximum cutoff energy of $\hbar\omega_{\text{max}} = I_p + 3.17U_p$, which scales **quadratically** with the laser wavelength and intensity [41]. Furthermore,

due to the half-cycle symmetry of the driving field in centrosymmetric media, the emitted spectrum consists exclusively of **odd harmonics** (see Appendix C.2).

3.3.2 The Solid-State Paradigm

In 2011, Ghimire et al. made the groundbreaking observation of HHG from a bulk ZnO crystal [42]. This launched a new field of attosecond condensed matter physics, revealing that the mechanisms in solids differ fundamentally from those in gases [9, 43]:

- **Delocalized States:** In a crystal, valence and conduction states are delocalized Bloch waves. An excited electron does not need to return to its original parent ion; it can recombine with a hole at any equivalent lattice site, making the process inherently coherent across the crystal.
- **Linear Cutoff Scaling:** In stark contrast to the quadratic scaling in gases, the high-energy cutoff in solids scales **linearly** with the peak electric field amplitude E_0 . This is a direct consequence of dynamical Bloch oscillations, where the maximum excursion in k -space is proportional to E_0 [9, 42].
- **Multiple Plateaus:** Solids possess multiple energy bands. Transitions between different pairs of valence and conduction bands can lead to distinct, secondary plateaus in the harmonic spectrum, encoding rich information about the material's band structure [44, 45].

3.3.3 The Basic Observable and Spectral Features

The fundamental macroscopic observable in solid-state HHG is the time-dependent current density $\mathbf{J}(t)$. According to Larmor's theorem, accelerating charges emit electromagnetic radiation. The emitted HHG spectrum $S(\omega)$ is proportional to the square of the Fourier transform of the time-derivative of the current (the acceleration):

$$S(\omega) \propto \omega^2 \left| \int_{-\infty}^{\infty} \mathbf{J}(t) e^{i\omega t} dt \right|^2 = \omega^2 |\mathbf{J}(\omega)|^2. \quad (3.3)$$

When plotted as a function of harmonic order, a typical solid-state HHG spectrum exhibits three distinct regimes [9, 10]:

1. **Perturbative Regime:** At low harmonic orders (typically below the minimum bandgap), the harmonic intensity follows a strict power law $I_{q\omega} \propto I_{\omega}^q$, characteristic of standard perturbative $\chi^{(q)}$ nonlinear optics.

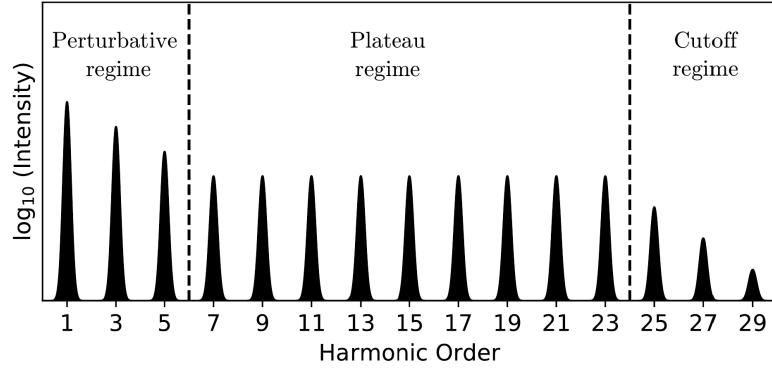


Figure 3.4: A typical high-order harmonic spectrum.

2. **Non-perturbative Plateau:** Above the bandgap, the harmonic intensities do not decay exponentially. Instead, they form a broad plateau of comparable intensity, signaling the onset of non-perturbative, strong-field dynamics.
3. **Cutoff:** The spectrum terminates in an abrupt high-energy cutoff. As established in the solid-state paradigm, this cutoff energy scales linearly with the peak electric field amplitude E_0 , reflecting the maximum excursion of the electron-hole pair in the Brillouin zone [42].

3.4 Interband and Intraband Dynamics

To understand the microscopic origin of solid-state HHG, the total current density $\mathbf{J}(t)$ must be decomposed into two distinct contributions: the intraband current and the interband current [9, 10].

$$\mathbf{J}(t) = \mathbf{J}_{\text{intra}}(t) + \mathbf{J}_{\text{inter}}(t). \quad (3.4)$$

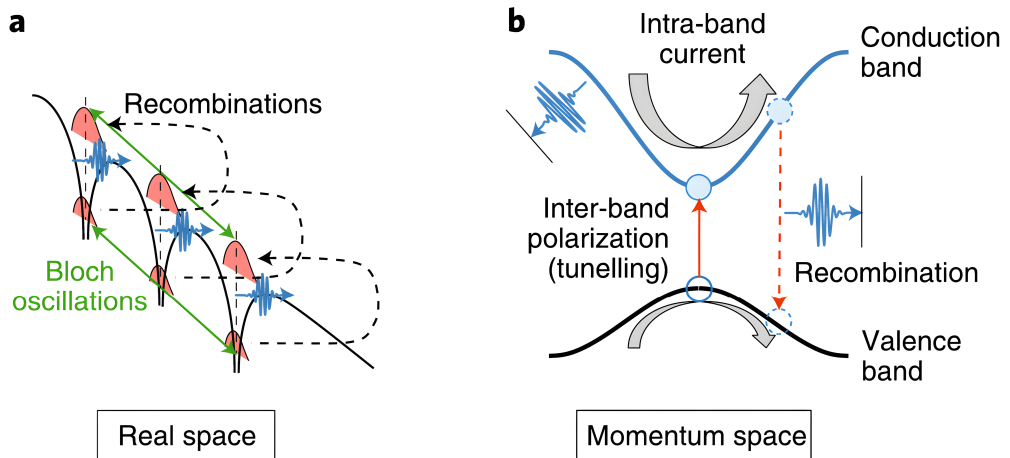


Figure 3.5: Microscopic mechanisms for solid-state HHG in (a) Real and (b) Momentum spaces.

3.4.1 Intraband Current and Anomalous Velocity

The intraband current arises from the acceleration of charge carriers (electrons in the conduction band and holes in the valence band) *within* their respective energy bands. In the semiclassical picture, the velocity of a wavepacket in band n is given by the group velocity $\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} \varepsilon_n(\mathbf{k})$. The intraband current is the sum of these velocities weighted by the band populations $\rho_{nn}(\mathbf{k}, t)$:

$$\mathbf{J}_{\text{intra}}(t) = -e \sum_n \int_{\text{BZ}} \frac{d^d k}{(2\pi)^d} \mathbf{v}_n(\mathbf{k}(t)) \rho_{nn}(\mathbf{k}, t), \quad (3.5)$$

where $\mathbf{k}(t) = \mathbf{k} - \frac{e}{\hbar} \mathbf{A}(t)$ is the crystal momentum shifted by the laser vector potential $\mathbf{A}(t)$ as derived via the Peierls substitution in Chapter 2. The nonlinearity in $\mathbf{J}_{\text{intra}}$ originates from the anharmonicity of the band structure $\varepsilon_n(\mathbf{k})$, leading to dynamical Bloch oscillations.

Topological Connection: As established in Chapter 1, the semiclassical equations of motion for a wavepacket include an **anomalous velocity** term proportional to the Berry curvature $\mathbf{\Omega}_n(\mathbf{k})$. The full velocity is $\dot{\mathbf{r}} = \mathbf{v}_n(\mathbf{k}) - \frac{e}{\hbar} \mathbf{E}(t) \times \mathbf{\Omega}_n(\mathbf{k})$. Consequently, the intraband current contains a direct contribution from the Berry curvature:

$$\mathbf{J}_{\text{anom}}(t) = \frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^d k}{(2\pi)^d} (\mathbf{E}(t) \times \mathbf{\Omega}_n(\mathbf{k}(t))) \rho_{nn}(\mathbf{k}, t). \quad (3.6)$$

This anomalous current flows transversely to the applied electric field and is a direct macroscopic manifestation of the band's topology.

3.4.2 Interband Current and the Berry Connection

The interband current arises from the coherent superposition of states across different bands, physically corresponding to the generation of electron-hole pairs and their subsequent recombination. It is given by the time derivative of the interband macroscopic polarization $\mathbf{P}_{\text{inter}}(t)$:

$$\mathbf{J}_{\text{inter}}(t) = \frac{d}{dt} \mathbf{P}_{\text{inter}}(t) = \frac{d}{dt} \left(-e \sum_{n \neq m} \int_{\text{BZ}} \frac{d^d k}{(2\pi)^d} \mathbf{d}_{nm}(\mathbf{k}(t)) \rho_{nm}(\mathbf{k}, t) \right) + \text{c.c.}, \quad (3.7)$$

where $\rho_{nm}(\mathbf{k}, t)$ is the interband coherence (polarization) between bands n and m .

The crucial quantity governing this process is the **transition dipole moment** $\mathbf{d}_{nm}(\mathbf{k})$, defined as:

$$\mathbf{d}_{nm}(\mathbf{k}) = i \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{m\mathbf{k}} \rangle. \quad (3.8)$$

Topological Connection: For the diagonal elements ($n = m$), the transition dipole moment is exactly the **Berry connection** $\mathcal{A}_n(\mathbf{k})$ defined in Eq. (1.6)! The interband HHG process is initiated by strong-field Landau-Zener tunneling, the probability of which is governed by the off-diagonal Berry connection. When the electron and hole recombine, the emitted photon carries a phase imprinted by the geometric phase accumulated during their excursion in the Brillouin zone.

3.5 Symmetry and Topological Fingerprints in HHG

The rich interplay between the laser field, the crystal lattice, and the band geometry means that the HHG spectrum is highly sensitive to the symmetries and topology of the material.

3.5.1 Selection Rules and Dynamical Symmetry Breaking

In a crystal possessing inversion symmetry (centrosymmetric), the band structure satisfies $\varepsilon_n(\mathbf{k}) = \varepsilon_n(-\mathbf{k})$, and the dipole matrix elements obey specific parity relations as rigorously derived in Appendix C.2. Under a linearly polarized driving field, the total current exhibits a half-cycle symmetry: $\mathbf{J}(t + T/2) = -\mathbf{J}(t)$. This symmetry constraint leads to the emission of **only odd harmonics**,

However, this selection rule can be broken dynamically. By utilizing a two-color driving field (e.g., mixing the fundamental frequency ω with its second harmonic 2ω) or by using circularly polarized light, the inversion symmetry of the total Hamiltonian is broken on a sub-cycle timescale. This dynamical symmetry breaking, as discussed in Section 2.6.1, allows even-order harmonics to appear, providing a sensitive interferometric probe of the crystal's structural and electronic symmetries [10, 37].

3.5.2 Topological Signatures in the Harmonic Spectrum

Because the microscopic currents $\mathbf{J}_{\text{intra}}$ and $\mathbf{J}_{\text{inter}}$ are fundamentally governed by the Berry curvature and Berry connection, HHG serves as an ultrafast, all-optical probe of topological invariants [9, 10]:

- **Phase Shifts and the Zak Phase:** In 1D systems, the phase of the transition dipole moment $\mathbf{d}_{cv}(k)$ is directly related to the Zak phase. When an electron-hole pair recombines, the emitted harmonic radiation carries a phase shift determined by the topological invariant. By interferometrically

measuring the phase of the emitted harmonics, one can directly extract the Zak phase and detect topological phase transitions [10].

- **Circular Dichroism and the Chern Number:** In 2D systems, the anomalous velocity term $\mathbf{J}_{\text{anom}}$ (Eq. (3.6)) generates a transverse current proportional to the integral of the Berry curvature (the Chern number). If the solid is driven by circularly polarized light, the emitted harmonic spectrum will exhibit a strong preference for a specific helicity, known as High-Harmonic Circular Dichroism (HHCD). The degree of this dichroism is directly linked to the Chern number of the occupied bands [10].
- **Band Structure Tomography:** The cutoff energy and the interference minima in the HHG spectrum encode the exact shape of the band structure and the Brillouin zone boundaries. By analyzing the polarization and intensity of the harmonics as a function of the laser polarization angle, one can reconstruct the full momentum-dependent Berry curvature and transition dipole phases, a technique known as band structure tomography [9].

3.5.3 Summary

In summary, High Harmonic Generation in solids transcends the perturbative limits of traditional nonlinear optics. The non-perturbative plateau and cutoff are governed by the ultrafast, sub-cycle dynamics of Bloch electrons. Crucially, the intraband and interband currents are inextricably linked to the geometric and topological properties of the band structure via the Berry curvature and Berry connection. In the following chapters, we will apply this theoretical framework to 1D/2D topological models (SSH, Rice-Mele, QWZ, Haldane), demonstrating how light-induced topological phase transitions manifest as distinct, measurable signatures in the high-harmonic spectrum.

Chapter 4: 1D Topological Insulators: Static and Driven Phases

4.1 Introduction

One-dimensional models occupy a special place in the study of topological phases: they are simple enough to be solved analytically in many limits, yet rich enough to exhibit all the essential features of bulk–boundary correspondence, symmetry-protected invariants, and light-induced phase transitions. They also serve as the natural testing ground for the theoretical machinery developed in the preceding chapters.

This chapter studies two complementary 1D models. The **Su-Schrieffer-Heeger (SSH) model**, originally introduced to describe soliton excitations in polyacetylene [?], is the minimal realization of a chiral-symmetry-protected topological insulator. Its topological invariant is the winding number, which is directly related to the Zak phase introduced in Section 1.6.1. The **Rice-Mele model** extends the SSH chain by adding a staggered on-site potential. This breaks chiral symmetry while preserving inversion symmetry, and the topological invariant shifts from a strict winding number to a quantized Zak phase tied to the macroscopic polarization through the modern theory of polarization.

For each model we proceed in three stages. First, we establish the static band structure and topological invariants, verifying the bulk–boundary correspondence through explicit edge-state calculations. Second, we apply the Floquet formalism of Chapter 2 to map out the light-induced topological phase diagrams as a function of driving amplitude. Third, we deploy the HHG framework of Chapter 3 to demonstrate that the even-to-odd harmonic ratio provides a direct, all-optical probe of both the static and Floquet topological invariants.

4.2 Static 1D Topological Models

4.2.1 The Su-Schrieffer-Heeger (SSH) Model

Lattice Structure and Bloch Hamiltonian

The SSH model describes spinless fermions on a dimerized chain of N unit cells, each hosting two sublattice sites A and B . The alternating hopping amplitudes — intracell t_1 and intercell t_2 — are the only free parameters. The real-space Hamiltonian is:

$$\hat{H}_{\text{SSH}} = t_1 \sum_{n=1}^N (|n_B\rangle\langle n_A| + \text{h.c.}) + t_2 \sum_{n=1}^{N-1} (|(n+1)_A\rangle\langle n_B| + \text{h.c.}). \quad (4.1)$$

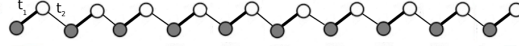


Figure 4.1: Schematic of the SSH model. Blue bonds denote intracell hopping t_1 ; red bonds denote intercell hopping t_2 .

Fourier-transforming to momentum space yields the 2×2 Bloch Hamiltonian:

$$H_{\text{SSH}}(k) = d_x(k) \sigma_x + d_y(k) \sigma_y, \quad (4.2)$$

with

$$d_x(k) = t_1 + t_2 \cos k, \quad d_y(k) = t_2 \sin k, \quad d_z(k) = 0. \quad (4.3)$$

The energy bands are $E_{\pm}(k) = \pm|\mathbf{d}(k)| = \pm\sqrt{t_1^2 + t_2^2 + 2t_1t_2 \cos k}$.

Symmetry Analysis and Topological Classification

The bipartite structure of the SSH chain enforces chiral (sublattice) symmetry with $\Gamma = \sigma_z$, which constrains $\mathbf{d}(k)$ strictly to the xy -plane ($d_z = 0$) for all k . As established in Section 1.6.1, this quantizes the Zak phase to 0 or π , and equivalently restricts the winding number $\nu \in \{0, 1\}$, which counts the number of times the trajectory of $\mathbf{d}(k)$ encircles the origin as k traverses the Brillouin zone:

- $t_1 > t_2$: the loop does not encircle the origin $\Rightarrow \nu = 0$ (trivial).
- $t_1 < t_2$: the loop encircles the origin once $\Rightarrow \nu = 1$ (topological).
- $t_1 = t_2$: the gap closes and the topological invariant is undefined.

The topological phase transition is therefore not driven by any symmetry breaking; both phases share the same symmetries. The two phases are distinguished solely by the global topology of the Bloch eigenstates over the Brillouin zone.

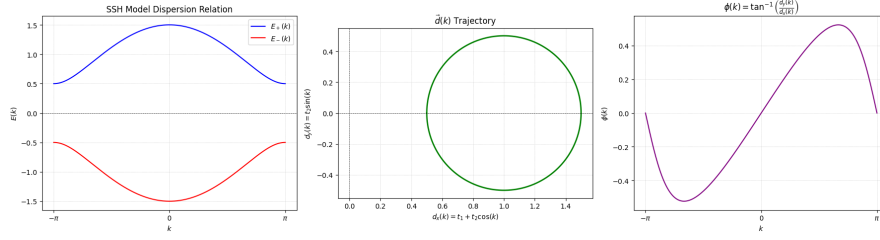


Figure 4.2: Bulk properties of the SSH model ($t_1 = 1.0$, $t_2 = 0.5$, trivial phase). (a) Dispersion relation. (b) $\mathbf{d}(k)$ trajectory in the d_x - d_y plane, which does not encircle the origin. (c) Phase $\phi(k)$ of the off-diagonal element.

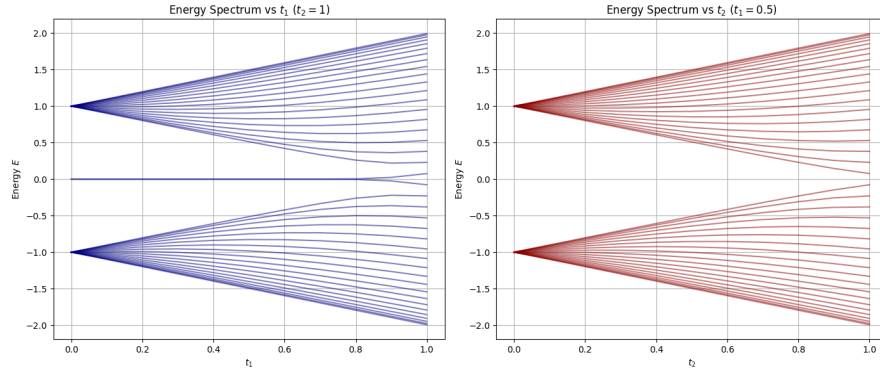


Figure 4.3: Energy gap of the SSH model as a function of hopping parameters. (a) vs t_1 (fixed $t_2 = 1.0$). (b) vs t_2 (fixed $t_1 = 0.5$). The gap closes at $t_1 = t_2$, marking the topological phase transition.

Effect of Symmetry Breaking: Precursor to the Rice-Mele Model

If we artificially introduce a staggered on-site potential $\Delta \neq 0$, the $\mathbf{d}$ -vector acquires a nonzero z -component $d_z = \Delta$. The trajectory of $\mathbf{d}(k)$ lifts off the xy -plane into a 3D loop. Chiral symmetry is broken, the winding number ceases to be a well-defined integer, and the Zak phase becomes a continuous function of Δ . This is precisely the physics of the Rice-Mele model, studied in Section 4.2.2.

Bulk–Boundary Correspondence and Edge States

According to the bulk–boundary correspondence (Section 1.7), a nontrivial winding number $\nu = 1$ guarantees the existence of zero-energy states localized at the boundaries of a finite open chain. In the topological phase ($t_2 > t_1$), these zero-energy edge states appear symmetrically: one exponentially localized on the

Case 2: Winding Paths with Sublattice Potential

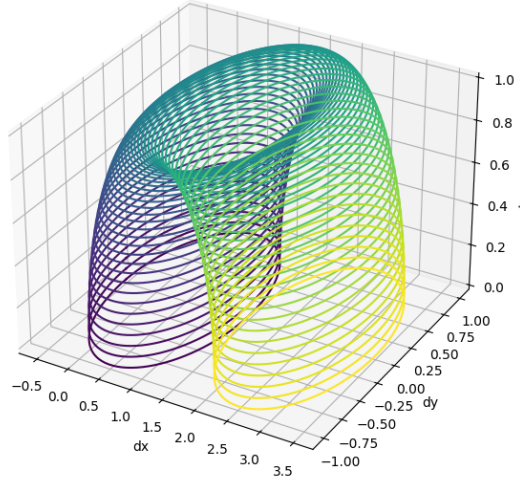
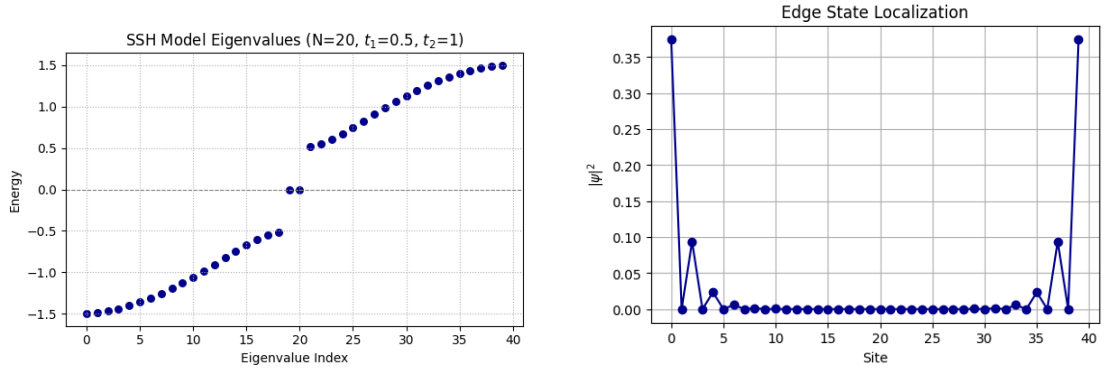


Figure 4.4: 3D trajectory of the $\mathbf{d}(k)$ vector when a staggered sublattice potential is introduced, illustrating the onset of chiral symmetry breaking.

A -sublattice at the left end, and one on the B -sublattice at the right end. Their energy is pinned to zero by chiral symmetry; they cannot be removed unless the gap closes or chiral symmetry is explicitly broken.



(a) Finite-chain spectrum ($N = 20$, $t_1 = 0.5$, $t_2 = 1.0$). Two states appear at $E = 0$ within the bulk gap.

(b) Probability density $|\psi|^2$ of the zero-energy states, showing exponential localization at the chain boundaries.

Figure 4.5: Bulk–boundary correspondence in the topological SSH phase. The two degenerate zero-energy edge states are a direct consequence of $\nu = 1$.

4.2.2 The Rice-Mele Model

The SSH model is an idealized limit: real 1D systems, as well as many engineered quantum platforms, inevitably possess inequivalent sublattice energies. The **Rice-**

Mele (RM) model accounts for this by adding a staggered on-site potential $\pm\Delta$:

$$\hat{H}_{\text{RM}} = \hat{H}_{\text{SSH}} + \Delta \sum_n (|n_A\rangle\langle n_A| - |n_B\rangle\langle n_B|). \quad (4.4)$$

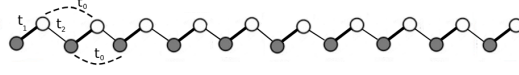


Figure 4.6: Schematic of the Rice-Mele model. In addition to the staggered hoppings t_1 and t_2 , each sublattice carries an on-site energy $\pm\Delta$.

In the Bloch basis, the Hamiltonian becomes:

$$H_{\text{RM}}(k) = \begin{pmatrix} \Delta & t_1 + t_2 e^{-ik} \\ t_1 + t_2 e^{ik} & -\Delta \end{pmatrix} = \mathbf{d}_{\text{RM}}(k) \cdot \boldsymbol{\sigma}, \quad (4.5)$$

with the three-component pseudospin vector:

$$\mathbf{d}_{\text{RM}}(k) = (t_1 + t_2 \cos k, t_2 \sin k, \Delta). \quad (4.6)$$

The z -component $d_z = \Delta$ is constant and independent of k . It lifts the $\mathbf{d}(k)$ trajectory off the equatorial plane, breaking chiral symmetry. The energy dispersion is $E_{\pm}(k) = \pm \sqrt{(t_1 + t_2 \cos k)^2 + t_2^2 \sin^2 k + \Delta^2}$, and the corresponding eigenstates are:

$$|\psi_{\pm}(k)\rangle = \begin{pmatrix} \pm e^{-i\phi(k)} \\ 1 \end{pmatrix}, \quad \phi(k) = \arctan\left(\frac{t_2 \sin k}{t_1 + t_2 \cos k}\right). \quad (4.7)$$

Symmetry Analysis

The staggered potential Δ breaks chiral symmetry, because the σ_z term commutes with $\Gamma = \sigma_z$ rather than anticommutes. Consequently, the spectrum is no longer symmetric about zero energy, and the strict winding number ν is ill-defined. However, the Rice-Mele model preserves **inversion symmetry** with $\mathcal{P} = \sigma_x$, since swapping the two sublattices and negating k leaves $H_{\text{RM}}(k)$ invariant. As shown in Section 1.6.1, inversion symmetry is sufficient to quantize the Zak phase to 0 or π even in the absence of chiral symmetry.

Modern Theory of Polarization

The topological Zak phase of the Rice-Mele model is directly related to the macroscopic electrical polarization through the modern theory of polarization [?, ?]:

$$P = -\frac{e}{2\pi} \int_{-\pi}^{\pi} \mathcal{A}(k) dk = \frac{e}{2\pi} \gamma_{\text{Zak}}, \quad (4.8)$$

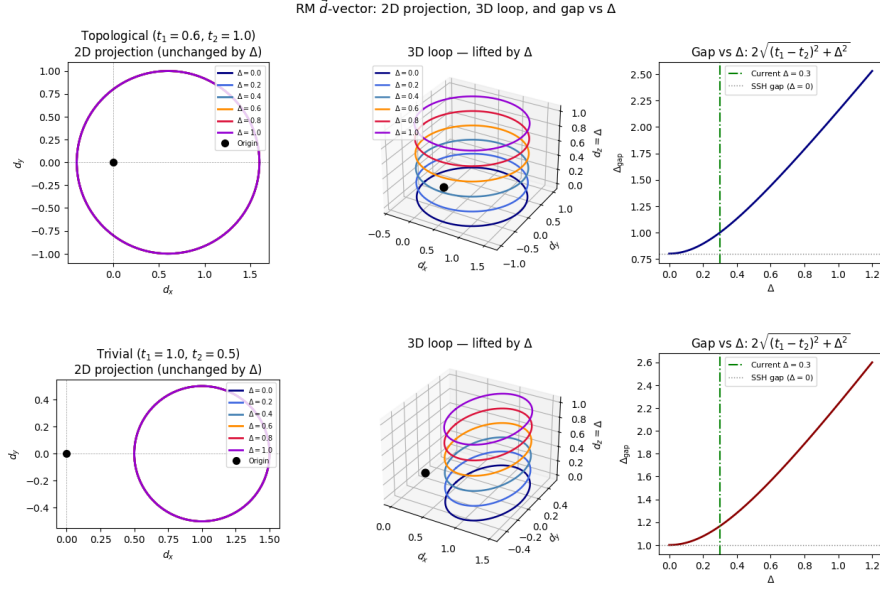


Figure 4.7: Rice-Mele $\vec{d}$ -vector visualization. (a) 2D projection showing the loop lifted off the equatorial plane. (b) Full 3D loop, where $d_z = \Delta$ controls the out-of-plane tilt. (c) Band gap as a function of Δ .

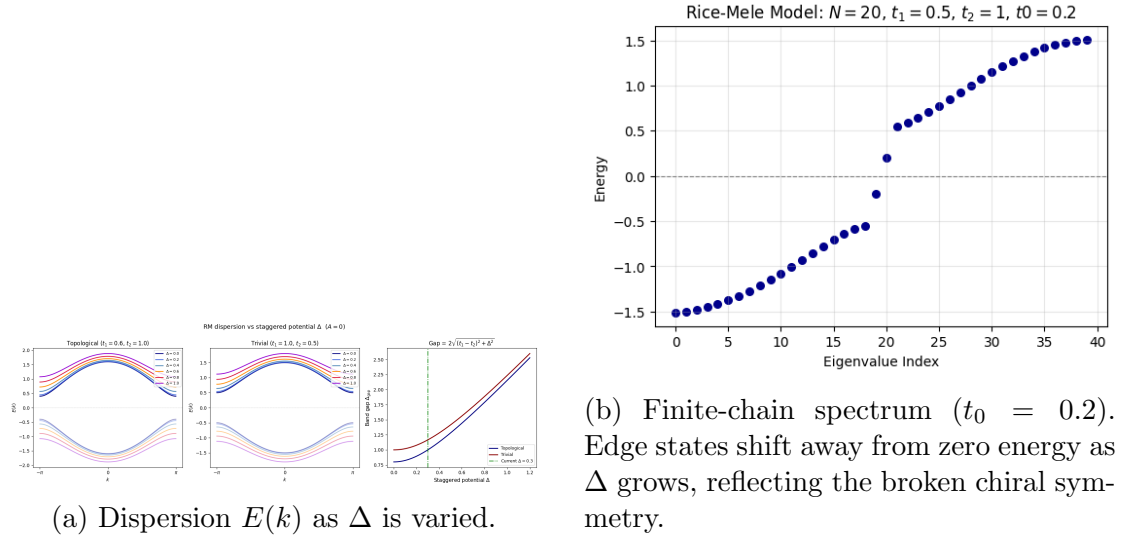


Figure 4.8: Effect of the staggered potential on the Rice-Mele spectrum.

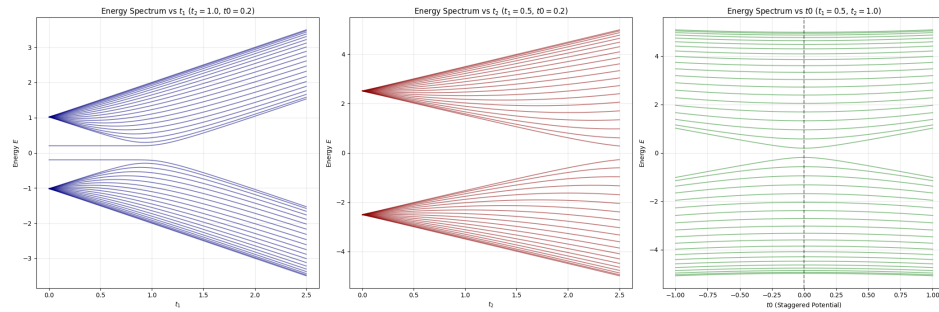


Figure 4.9: Energy spectrum of the Rice-Mele model as a function of (a) t_1 , (b) t_2 , and (c) staggered potential Δ . The dashed vertical line in (c) marks $\Delta = 0$ (SSH limit). Finite Δ shifts the edge states into the bulk gap without closing it.

where $\mathcal{A}(k) = i\langle u_k | \partial_k u_k \rangle$ is the Berry connection of the occupied band. For the lower band of the Rice-Mele model, the phase of the off-diagonal element $\phi(k)$ (Eq. (4.7)) provides a direct geometric handle on the polarization:

$$P = \frac{e}{4\pi} \Delta\phi(k) \Big|_{-\pi}^{\pi}. \quad (4.9)$$

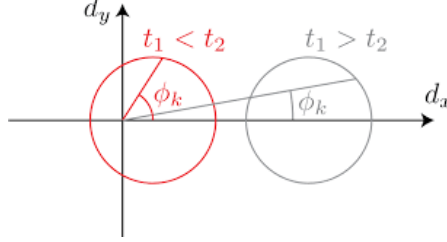
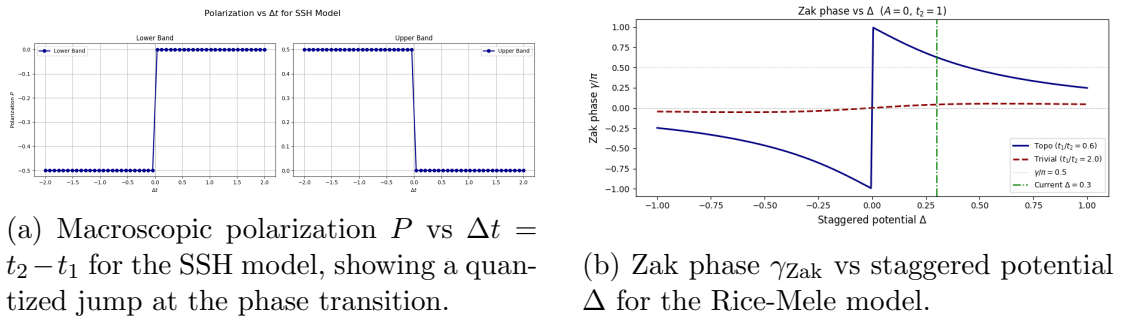


Figure 4.10: Polar angle $\phi(k)$ of the off-diagonal Hamiltonian element across the Brillouin zone. Its total winding determines the Zak phase and hence the macroscopic polarization.

The polarization is quantized to one of two values:

- **Trivial phase** ($t_1 > t_2$): The xy -projection of $\mathbf{d}(k)$ does not encircle the origin, so $\Delta\phi = 0$ and $P = 0$.
- **Topological phase** ($t_1 < t_2$): The xy -projection encircles the origin, so $\Delta\phi = 2\pi$ and $P = e/2 \pmod{e}$.

The topological phase transition thus corresponds to a discontinuous jump in the macroscopic polarization by $e/2$, providing a direct connection between the microscopic band topology and a macroscopic observable. This quantized pumping can be realized dynamically by adiabatically cycling through the RM parameter space — the celebrated Thouless pump [?].



(a) Macroscopic polarization P vs $\Delta t = t_2 - t_1$ for the SSH model, showing a quantized jump at the phase transition.

(b) Zak phase γ_{Zak} vs staggered potential Δ for the Rice-Mele model.

Figure 4.11: Topological invariants and macroscopic polarization across phase transitions in the 1D models.

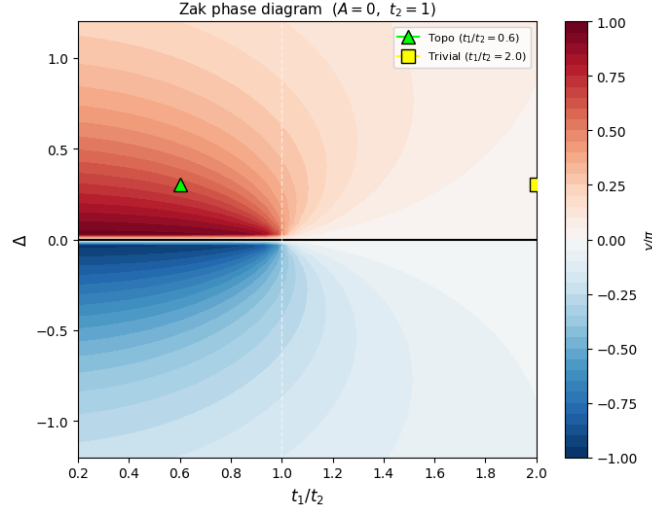


Figure 4.12: Topological phase diagram of the Rice-Mele model in the (t_1, Δ) parameter space (fixed $t_2 = 1$). The phase boundary at $t_1 = t_2$ separates trivial ($P = 0$) and topological ($P = e/2$) regions. Finite Δ never closes the gap; it merely shifts the edge states away from zero energy.

4.3 Floquet Engineering of 1D Topological Phases

The theoretical framework developed in Chapter 2 showed that a periodically driven system is governed by an effective Floquet Hamiltonian H_F whose topological invariants can differ from those of the static system. Here we apply this framework to the SSH and Rice-Mele models, demonstrating that a mid-infrared laser field can drive transitions between topologically distinct Floquet phases even when the static parameters are held fixed.

4.3.1 Laser Parameters and the Peierls Substitution

All simulations employ the spatially uniform vector potential $A(t) = (F_0/\omega) \cos(\omega t)$, with the following parameters chosen to reflect experimental mid-infrared solid-state HHG conditions [42]:

- Angular frequency: $\omega = 0.014$ a.u. ($\lambda = 3.2 \mu\text{m}$)
- Peak electric field: $F_0 = 0.0045$ a.u. ($I \approx 0.8 \text{ TW/cm}^2$)
- Pulse duration: 8 optical cycles (~ 85 fs); sine-squared envelope

Via the Peierls substitution (Section 2.2), the hopping amplitudes acquire time-dependent phases: a bond of length δ aligned along $\hat{x}$ transforms as $t \rightarrow t e^{-iA(t)\delta}$. For the SSH chain, this modifies the off-diagonal Bloch element to:

$$h(k, t) = t_1 + t_2 e^{-i[k+A(t)]}. \quad (4.10)$$

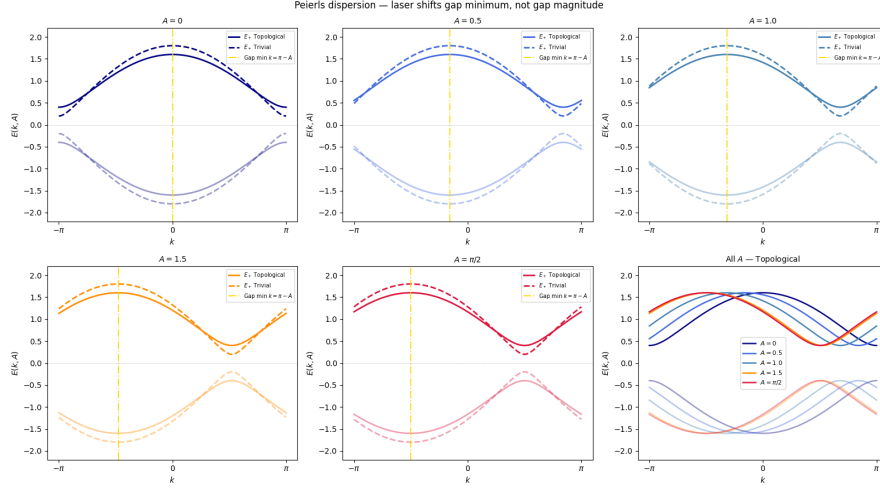


Figure 4.13: Instantaneous band structure of the SSH model under laser driving. The Peierls substitution shifts the momentum-space gap minimum from $k = \pi$ to $k = \pi - A(t)$, but the instantaneous gap $2|t_1 - t_2|$ is independent of $A(t)$. Topological transitions are therefore encoded in the Floquet spectrum, not the instantaneous one.

A crucial observation follows from the driven dispersion $E_{\pm}(k, t) = \pm\sqrt{t_1^2 + t_2^2 + 2t_1t_2 \cos(k + A(t))}$: the instantaneous gap $\Delta_{\text{inst}} = 2|t_1 - t_2|$ is entirely independent of $A(t)$. The laser never closes the instantaneous gap; it only slides the gap minimum through the Brillouin zone. Topological transitions in the driven system are therefore genuinely non-equilibrium phenomena, encoded in the Floquet quasi-energy spectrum rather than in any instantaneous snapshot of the band structure.

4.3.2 The Periodically Driven SSH Model

The one-cycle Floquet propagator is computed numerically via time-ordered integration:

$$U_F(k) = \mathcal{T} \exp\left(-i \int_0^T H_{\text{SSH}}(k, t) dt\right). \quad (4.11)$$

The quasi-energies $\varepsilon_{\pm}(k)$ are extracted from the eigenvalues $e^{-i\varepsilon_{\pm}T}$ of $U_F(k)$, and are confined to the Floquet-Brillouin zone $\varepsilon \in [-\pi/T, \pi/T)$.

The topology of the driven SSH system is characterized by the Floquet winding number ν_F , computed from the off-diagonal element $U_F^{01}(k)$ of the propagator:

$$\nu_F = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk \frac{\partial}{\partial k} \arg[U_F^{01}(k)]. \quad (4.12)$$

Gap closings in the quasi-energy spectrum — at $\varepsilon = 0$ or at the Floquet zone boundary $\varepsilon = \pm\pi/T$ — signal changes in ν_F and hence laser-induced topological phase transitions.

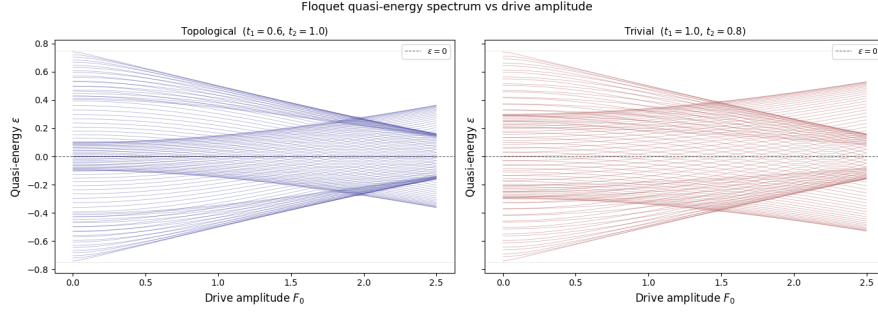
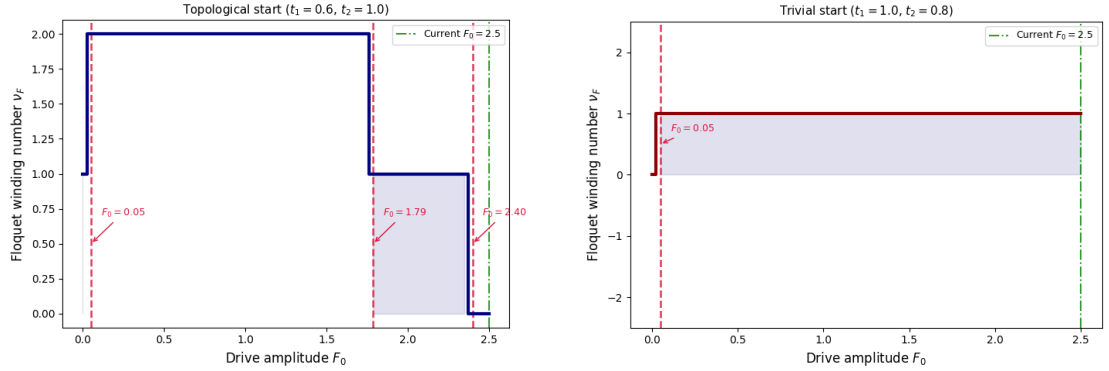


Figure 4.14: Floquet quasi-energy spectrum of the SSH model as a function of drive amplitude F_0 . Gap closings at $\varepsilon = 0$ (solid horizontal line) and $\varepsilon = \pm\pi/T$ (dashed lines) mark the boundaries of distinct Floquet topological phases.



(a) Starting in the topological static phase ($t_1 = 0.5, t_2 = 1.0$). (b) Starting in the trivial static phase ($t_1 = 1.0, t_2 = 0.5$).

Figure 4.15: Floquet winding number ν_F as a function of driving amplitude F_0 . In both cases, increasing the drive can push the system into Floquet phases with $\nu_F = 0, 1, 2$, demonstrating that periodic driving can both create and destroy topological order.

4.3.3 The Periodically Driven Rice-Mele Model

For the Rice-Mele model, the Bloch Hamiltonian under driving is:

$$H_{\text{RM}}(k, t) = \begin{pmatrix} \Delta & h^*(k, t) \\ h(k, t) & -\Delta \end{pmatrix}, \quad (4.13)$$

where $h(k, t) = t_1 + t_2 e^{-i[k+A(t)]}$ as before, but now $d_z = \Delta$ is a constant nonzero background. The instantaneous gap is enhanced to $\Delta_{\text{gap}} = 2\sqrt{(t_1 - t_2)^2 + \Delta^2}$.

Because chiral symmetry is absent, the relevant Floquet invariant is the **Floquet Zak phase** γ_F , defined as the Berry phase of the lower quasi-energy eigenstate across the Brillouin zone:

$$\gamma_F = -\text{Im} \ln \prod_{n=0}^{N_k-1} \langle u_{F,-}(k_n) | u_{F,-}(k_{n+1}) \rangle. \quad (4.14)$$

Jumps in γ_F (between 0 and π) occur when the Floquet quasi-energy gap closes, marking laser-induced topological transitions in the polar insulator. The Δ term generically broadens the phase boundaries relative to the SSH model, since a larger instantaneous gap requires a stronger drive to close.

4.4 High Harmonic Generation in 1D Models

4.4.1 Observables and Numerical Approach

The primary observable is the macroscopic current $J(t)$, computed as the expectation value of the Peierls-derived current operator (Section 2.2.3):

$$J(t) = \langle \Psi(t) | \hat{J}(t) | \Psi(t) \rangle, \quad \hat{J}(t) = -\frac{\partial H(k, A(t))}{\partial A}. \quad (4.15)$$

The time evolution is computed by integrating the time-dependent Schrödinger equation with a 4th-order Runge-Kutta scheme, initializing in the half-filled ground state. The HHG power spectrum is:

$$S(\Omega) = \left| \int_0^{t_{\text{max}}} J(t) e^{i\Omega t} dt \right|^2. \quad (4.16)$$

As established in Chapter 3, the total current decomposes into intraband and interband contributions. In 1D, the intraband component drives dynamical Bloch oscillations via the anharmonic band dispersion, while the interband component encodes the electron-hole recombination amplitude governed by the Berry connec-

tion. The HHG cutoff energy scales linearly with the peak field F_0 , in contrast to the quadratic scaling of gas-phase HHG.

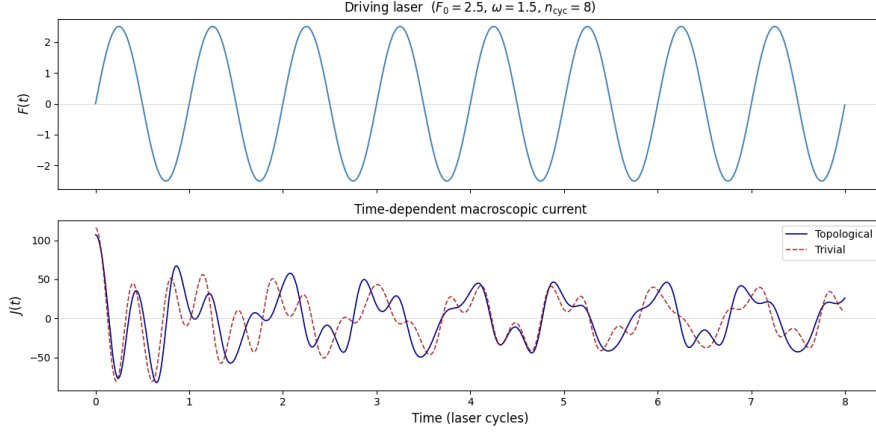


Figure 4.16: Time-dependent macroscopic current $J(t)$ (blue) and driving field $F(t)$ (red, scaled) for the SSH model. The current exhibits rich nonlinear dynamics, including high-frequency oscillations that give rise to the harmonic spectrum.

4.4.2 Topological Fingerprints in the Static HHG Spectrum

Figure 4.17 shows the HHG spectra for the static SSH model in both phases. Both spectra exhibit the three characteristic regimes identified in Section 3.3: a perturbative decay at low harmonic orders, a non-perturbative plateau above the band gap, and an abrupt cutoff.

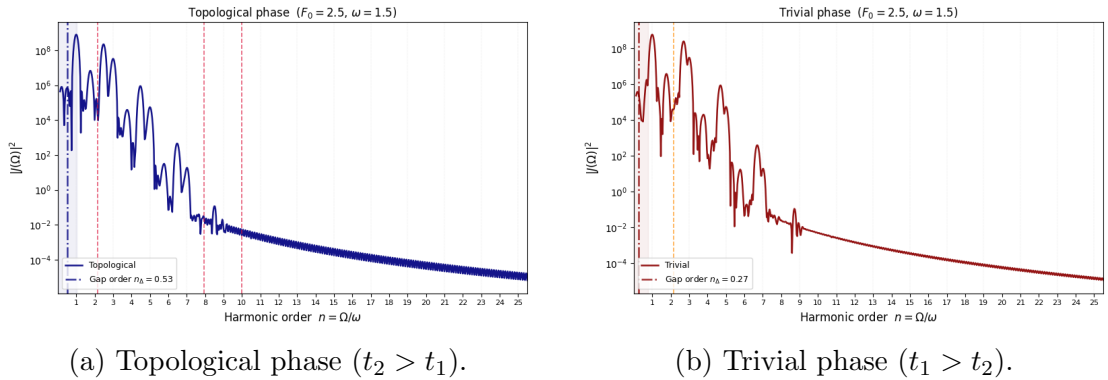
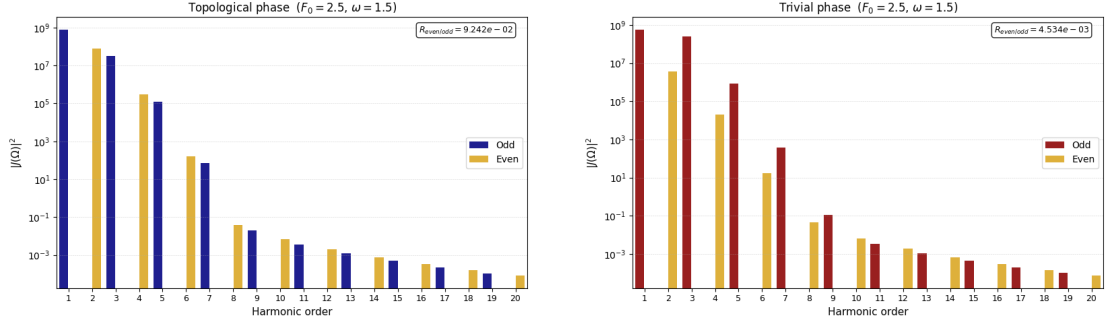


Figure 4.17: HHG power spectra for the static SSH model. Vertical dashed lines indicate the gap harmonic order $n_{\text{gap}} \approx \Delta_{\text{gap}}/\omega$.

The most striking topological signature lies in the **selection rules**. A bulk centrosymmetric system driven by a linearly polarized field satisfies the half-cycle symmetry $J(t + T/2) = -J(t)$, which, as shown in Appendix C.2, strictly forbids even harmonics. In the topological phase, however, the zero-energy edge states

localize at the chain boundaries and break this half-cycle symmetry locally. Their contribution to the time-dependent current generates a finite even-harmonic yield that is absent in the trivial phase.



(a) Topological phase: significant even-harmonic yield.

(b) Trivial phase: even harmonics suppressed.

Figure 4.18: Odd and even harmonic intensities for the two SSH phases. The even-to-odd ratio $R = \sum S_{\text{even}} / \sum S_{\text{odd}}$ is substantially higher in the topological phase, providing an all-optical probe of the bulk invariant via the bulk–boundary correspondence.

This observation establishes a direct connection between the bulk topological invariant and an experimentally measurable quantity: the even-to-odd harmonic ratio $R = \sum S_{\text{even}} / \sum S_{\text{odd}}$, which serves as an all-optical order parameter for the topological phase transition.

For the Rice-Mele model, the staggered potential Δ shifts the edge states away from zero energy, but they remain within the bulk gap. Their continued presence maintains an enhanced even-harmonic yield in the topological phase, confirming that the HHG signature is robust to chiral-symmetry breaking as long as inversion symmetry is preserved.

4.4.3 HHG as a Probe of Floquet Topological Transitions

We now turn to the driven system, where the laser field simultaneously probes and modifies the topology. As F_0 increases, the system undergoes the Floquet topological transitions identified in Section 4.3.2. Figure 4.19 plots $R(F_0)$ for both the topological and trivial initial SSH configurations.

The ratio $R(F_0)$ exhibits sharp, non-monotonic features precisely at the critical amplitudes F_0^* where ν_F changes. The physical mechanism is clear: at a Floquet transition, the quasi-energy gap closes and the Floquet edge states are rearranged, fundamentally altering the symmetry properties of the sub-cycle current and abruptly modifying the even-harmonic yield.

These results demonstrate that HHG is not merely a passive spectroscopic probe of static topology — it is an ultrafast tool capable of resolving *light-induced*

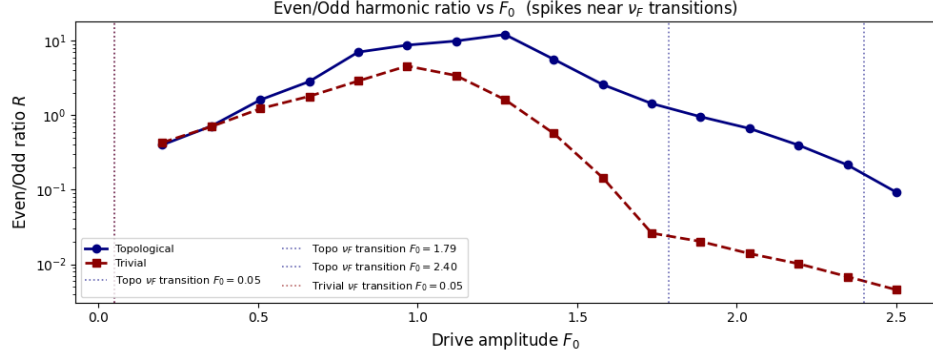


Figure 4.19: Even-to-odd harmonic ratio R as a function of driving amplitude F_0 for the SSH model. Vertical dashed lines mark the critical amplitudes F_0^* at which the Floquet winding number ν_F changes. The sharp features in R track the topological transitions with high fidelity.

topological phase transitions in real-time, with the even-to-odd harmonic ratio providing a clear, experimentally accessible signal.

4.4.4 HHG in the Rice-Mele Model

The Rice-Mele model displays qualitatively similar HHG signatures, with the presence of Δ leading to modified cutoff energies and harmonic yields compared to the SSH model. Figures 4.21 and 4.22 show the corresponding spectra, and Fig. 4.23 tracks the even-to-odd ratio as a function of F_0 .

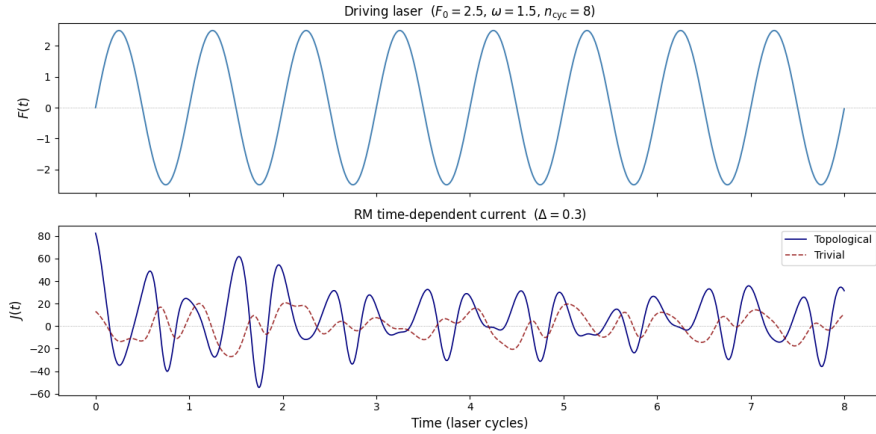


Figure 4.20: Macroscopic current for the Rice-Mele model ($\Delta = 0.2$). The nonlinear dynamics are qualitatively similar to the SSH case, with the staggered potential modifying the amplitude and phase of the oscillations.

The persistence of topological HHG signatures in the Rice-Mele model is significant: it establishes that the even-to-odd ratio is a robust indicator of inversion-symmetry-protected topology, not merely an artifact of the idealized chiral-symmetric SSH limit.

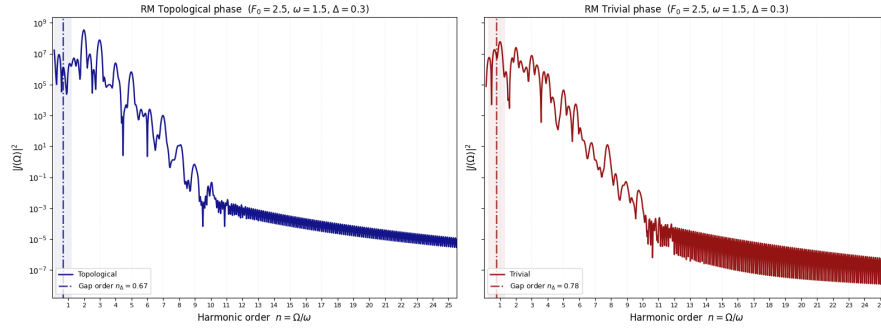


Figure 4.21: HHG spectra for the Rice-Mele model in the topological and trivial phases. The nonzero Δ increases the minimum gap, shifting the onset of the non-perturbative plateau to higher harmonic orders.

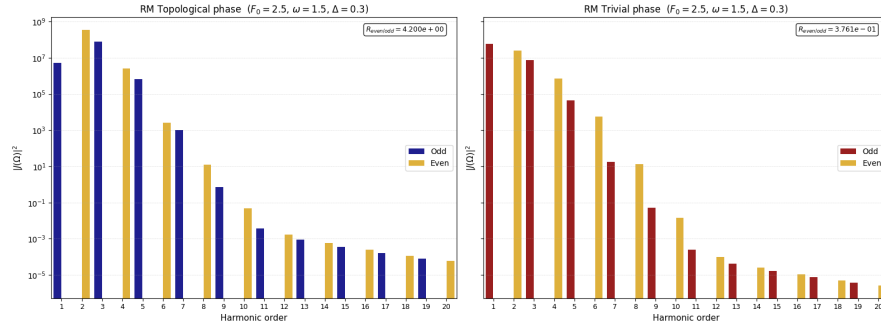


Figure 4.22: Odd and even harmonic intensities for the Rice-Mele model. Despite the broken chiral symmetry, the topological phase retains an enhanced even-harmonic yield, confirming that inversion symmetry alone is sufficient to support the topological HHG signature.

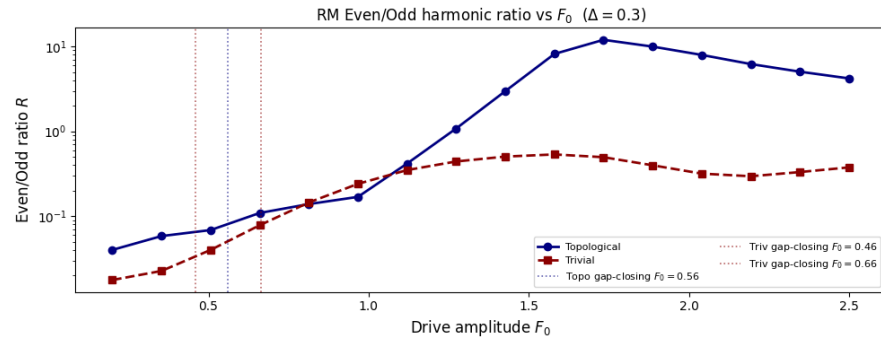


Figure 4.23: Even-to-odd ratio vs driving amplitude for the Rice-Mele model. Sharp features track the Floquet Zak phase transitions, demonstrating that HHG can probe topology beyond the chiral-symmetric SSH limit.

Chapter 5: 2D Topological Insulators: Static and Driven Phases

5.1 Introduction

Moving from one to two dimensions introduces a qualitatively new class of topological invariants. In 2D, the Berry curvature $\Omega(\mathbf{k})$ is a pseudoscalar field distributed over the Brillouin zone torus, and its integral over the entire zone yields the Chern number $\mathcal{C} \in \mathbb{Z}$ (Section 1.6.2). A nonzero Chern number implies the presence of chiral edge modes — unidirectional conducting channels that are immune to backscattering — and a quantized anomalous Hall conductance $\sigma_{xy} = \mathcal{C} e^2/h$, even without an applied magnetic field.

This chapter studies two paradigmatic 2D Chern insulators. The **Qi-Wu-Zhang (QWZ) model** provides the minimal lattice realization of a Chern insulator on a square lattice, and has the virtue of being directly related to the Rice-Mele Thouless pump by dimensional extension. The **Haldane model** [46] is defined on a honeycomb lattice and achieves a nonzero Chern number through complex next-nearest-neighbor hopping that breaks time-reversal symmetry without requiring net magnetic flux. Together, these two models capture the full range of phenomena relevant to 2D topological phases: integer Chern numbers, Dirac cones, valley-selective physics, and the interplay between sublattice and time-reversal symmetry breaking.

We follow the same three-stage strategy as Chapter 4: static properties and bulk–boundary correspondence, followed by Floquet engineering with circularly polarized light, and finally the HHG response as an all-optical probe of the driven topological phases.

5.2 Static 2D Topological Models

5.2.1 The Qi-Wu-Zhang (QWZ) Model

Bloch Hamiltonian and Band Structure

The QWZ model is constructed by replacing the cyclic time parameter of the 1D Rice-Mele pump with a second momentum coordinate k_y [47]. The resulting Bloch Hamiltonian takes the standard two-band form $H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$, with:

$$\mathbf{d}(\mathbf{k}) = \begin{pmatrix} \sin k_x \\ \sin k_y \\ t_0 + \cos k_x + \cos k_y \end{pmatrix}, \quad (5.1)$$

where t_0 is a tunable dimensionless mass parameter. The energy spectrum $E_{\pm}(\mathbf{k}) = \pm|\mathbf{d}(\mathbf{k})|$ is gapped everywhere in the Brillouin zone except at the isolated points where $\mathbf{d}(\mathbf{k}) = \mathbf{0}$.

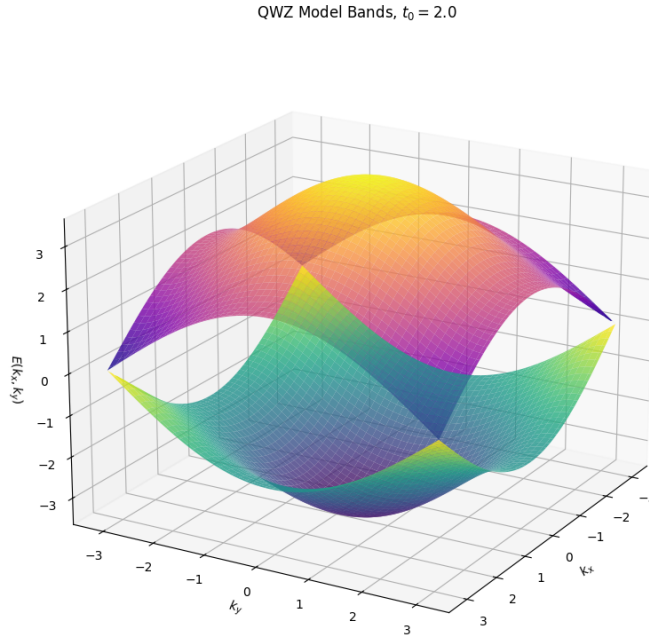


Figure 5.1: Energy bands of the QWZ model for $t_0 = 2.0$ (trivial phase). (a) Conduction band $E_+(\mathbf{k})$. (b) Valence band $E_-(\mathbf{k})$.

Topological Phase Diagram

The bulk gap closes when $\mathbf{d}(\mathbf{k}) = \mathbf{0}$. Because $d_x = \sin k_x$ and $d_y = \sin k_y$ both vanish only at the high-symmetry points $\{\Gamma, X, M\}$, gap closings are confined to

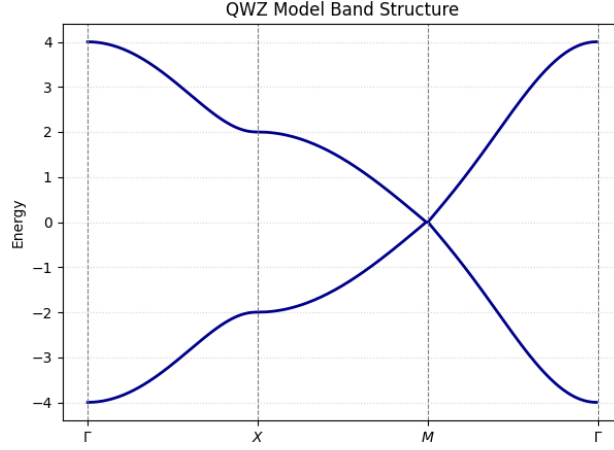


Figure 5.2: QWZ band structure along the high-symmetry path $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$ for $t_0 = 2.0$. The gap remains open throughout.

these isolated momenta. At each point, the closure condition $d_z = 0$ fixes the critical t_0 :

- $\Gamma = (0, 0)$: $d_z = t_0 + 2$; gap closes at $t_0 = -2$.
- $X = (0, \pi)$ or $(\pi, 0)$: $d_z = t_0$; gap closes at $t_0 = 0$.
- $M = (\pi, \pi)$: $d_z = t_0 - 2$; gap closes at $t_0 = +2$.

At each gap closing, a Dirac cone forms and the winding of $\hat{\mathbf{d}}(\mathbf{k})$ over the Brillouin zone torus changes by ± 1 . The resulting Chern number of the lower band is:

$$\mathcal{C} = \begin{cases} 0, & t_0 < -2 \text{ or } t_0 > 2, \\ -1, & -2 < t_0 < 0, \\ +1, & 0 < t_0 < 2. \end{cases} \quad (5.2)$$

Chiral Edge States and Bulk–Boundary Correspondence

To verify the bulk–boundary correspondence, we place the model in a ribbon geometry: N_x sites with open boundaries along x and periodic boundary conditions (conserved k_y) along y . In the topological phase ($|\mathcal{C}| = 1$), the ribbon spectrum reveals chiral edge states that traverse the bulk gap as a function of k_y . These modes are localized on opposite edges, propagate in opposite directions, and cannot hybridize or backscatter as long as the bulk gap remains open.

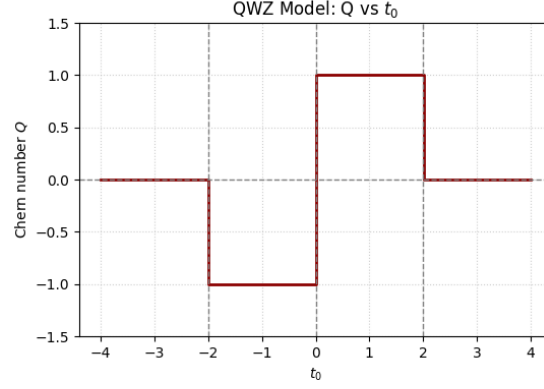


Figure 5.3: Chern number $\mathcal{C}$ of the lower QWZ band as a function of the mass parameter t_0 . Quantized plateaus are separated by gap closings at $t_0 = -2, 0, +2$.

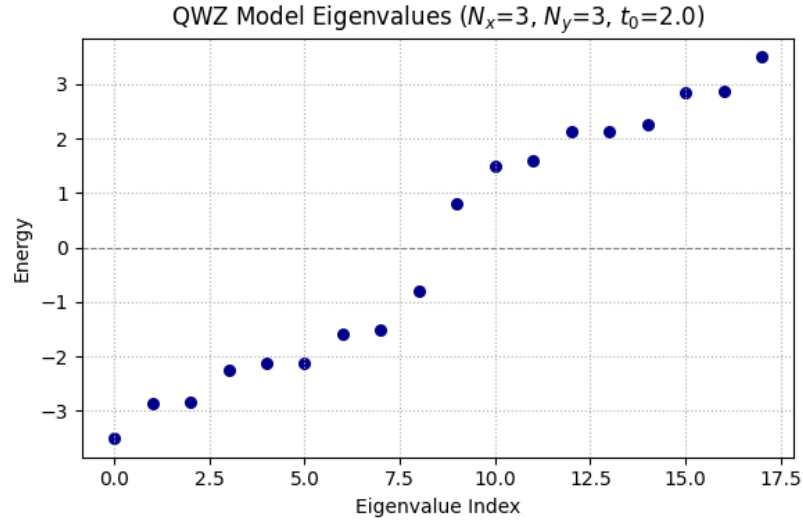


Figure 5.4: Ribbon spectrum of the QWZ model ($N_x = 20$, $t_0 = -1.5$, $\mathcal{C} = -1$) as a function of k_y . One chiral edge mode traverses the gap in each direction, consistent with $|\mathcal{C}| = 1$.

5.2.2 The Haldane Model

The Haldane model [46] demonstrates that a Chern insulating phase can be realized on the honeycomb lattice — the same lattice as graphene — using only local, flux-free interactions. Three ingredients are combined:

1. Nearest-neighbor hopping t_1 , which produces two Dirac cones at the K and K' points of the hexagonal Brillouin zone.
2. Complex next-nearest-neighbor hopping $t_2 e^{\pm i\phi}$ with a Haldane phase ϕ . This breaks time-reversal symmetry and opens a topological gap at the Dirac points.
3. A staggered sublattice potential t_0 (Semenoff mass), which also gaps the Dirac cones but breaks inversion symmetry, driving the system trivial.

The competition between these two gap-opening mechanisms — the topological $t_2 \sin \phi$ term and the trivial t_0 term — generates a rich phase diagram.

Bloch Hamiltonian

In the sublattice basis, $H_H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ with:

$$d_x(\mathbf{k}) = -t_1 \sum_{j=1}^3 \cos(\mathbf{k} \cdot \mathbf{a}_j), \quad (5.3)$$

$$d_y(\mathbf{k}) = -t_1 \sum_{j=1}^3 \sin(\mathbf{k} \cdot \mathbf{a}_j), \quad (5.4)$$

$$d_z(\mathbf{k}) = t_0 - 2t_2 \sin \phi \sum_{j=1}^3 \sin(\mathbf{k} \cdot \mathbf{b}_j), \quad (5.5)$$

where $\mathbf{a}_j$ and $\mathbf{b}_j$ are the nearest- and next-nearest-neighbor lattice vectors, respectively. The eigenstates are:

$$|\psi_{\pm}(\mathbf{k})\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{1 \pm d_z/|\mathbf{d}|} \\ \frac{d_x + id_y}{\sqrt{d_x^2 + d_y^2}} \sqrt{1 \mp d_z/|\mathbf{d}|} \end{pmatrix}. \quad (5.6)$$

Topological Phase Diagram

Gap closings are confined to the Dirac points, where $d_x = d_y = 0$ automatically. The effective masses at K and K' are:

$$d_z(K) = t_0 - 3\sqrt{3} t_2 \sin \phi, \quad d_z(K') = t_0 + 3\sqrt{3} t_2 \sin \phi. \quad (5.7)$$

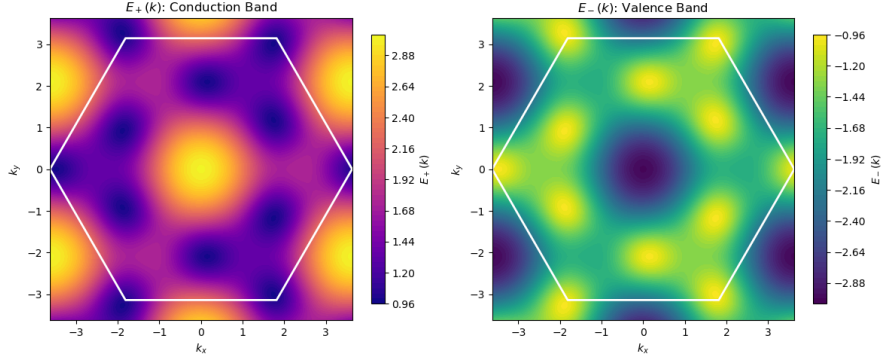


Figure 5.5: Haldane model band structure ($t_1 = 1.0$, $t_2 = 0.3$, $t_0 = 0.2$, $\phi = \pi/2$). (a) Conduction band. (b) Valence band. The white hexagon marks the first Brillouin zone of the honeycomb lattice.

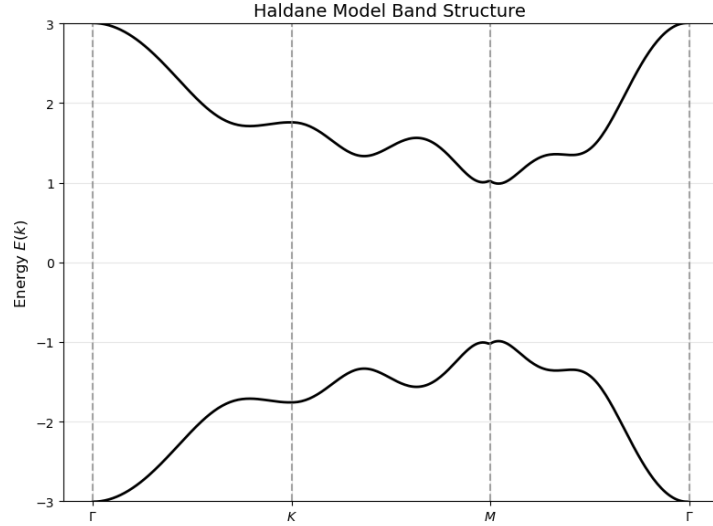


Figure 5.6: Haldane band structure along $\Gamma \rightarrow K \rightarrow M \rightarrow \Gamma$. The gap is opened at the K and K' Dirac points by the combination of t_0 and $t_2 \sin \phi$.

A gap closing at K (or K') occurs when the corresponding mass changes sign. Each such event changes the Chern number by ± 1 , generating the well-known $(t_0/t_2, \phi)$ phase diagram with regions of $\mathcal{C} = -1, 0, +1$.

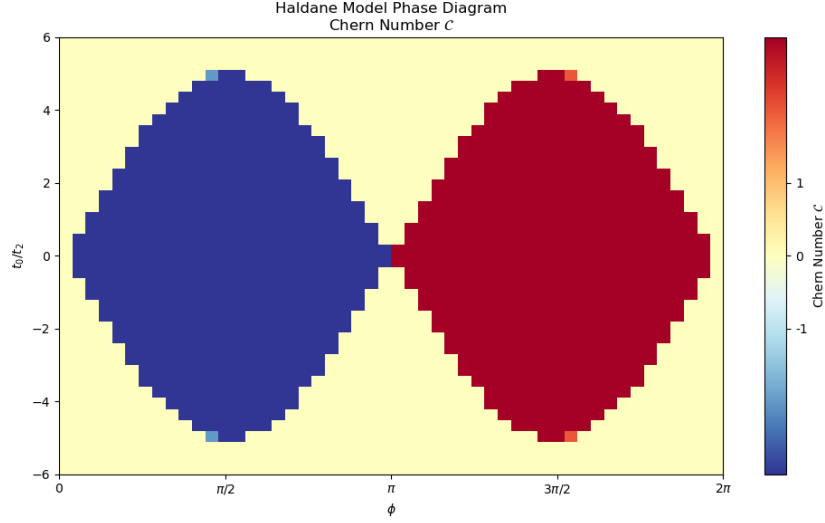


Figure 5.7: Chern number phase diagram of the Haldane model in the $(\phi, t_0/t_2)$ plane. The colored lobes correspond to $\mathcal{C} = \pm 1$; the white regions have $\mathcal{C} = 0$. Phase boundaries are set by $t_0 = \pm 3\sqrt{3} t_2 \sin \phi$.

Chiral Edge States

In the topological phase, the ribbon spectrum reveals chiral edge modes crossing the bulk gap. Unlike the helical edge states of $\mathbb{Z}_2$ topological insulators, these modes are strictly unidirectional: no time-reversal partner exists to allow backscattering.

5.3 Periodically Driven 2D Models

5.3.1 Circularly Polarized Drive and the Peierls Substitution

To engineer topological transitions in 2D, we employ a **circularly polarized** laser field. The vector potential traces a circle in the (A_x, A_y) plane:

$$A_x(t) = \frac{F_0}{\omega} \cos(\omega t), \quad A_y(t) = \frac{F_0}{\omega} \sin(\omega t), \quad (5.8)$$

using the same mid-infrared parameters as Chapter 4 ($F_0 = 2.5$ in natural units, $\hbar\omega = 1.5$). The Peierls substitution $\mathbf{k} \rightarrow \mathbf{k} + \mathbf{A}(t)$ simultaneously shifts the crystal

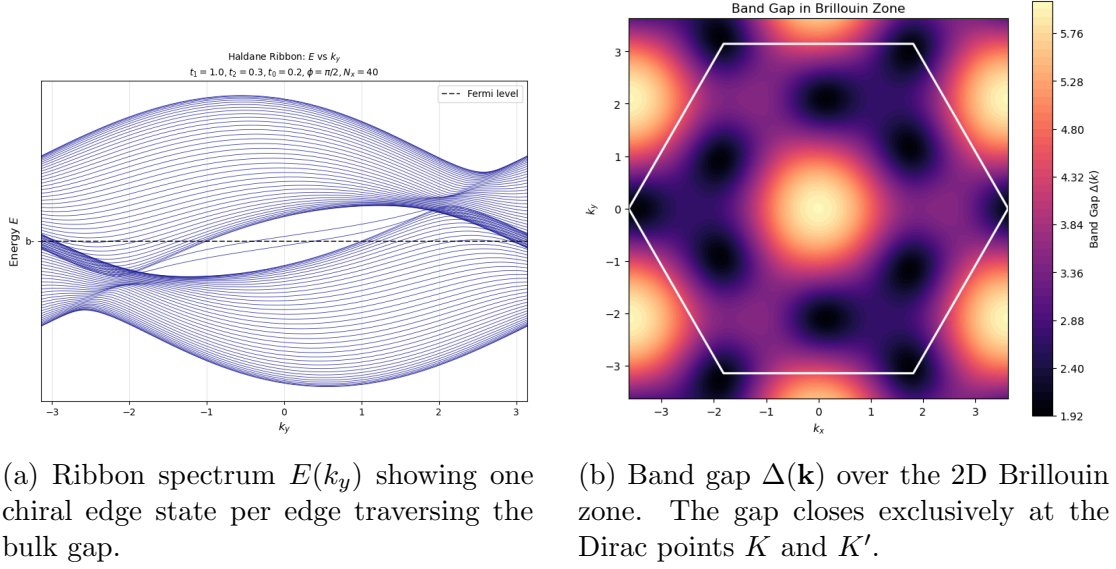


Figure 5.8: Bulk–boundary correspondence in the Haldane model ($t_1 = 1.0, t_2 = 0.1, t_0 = 0.3, \phi = \pi/2$).

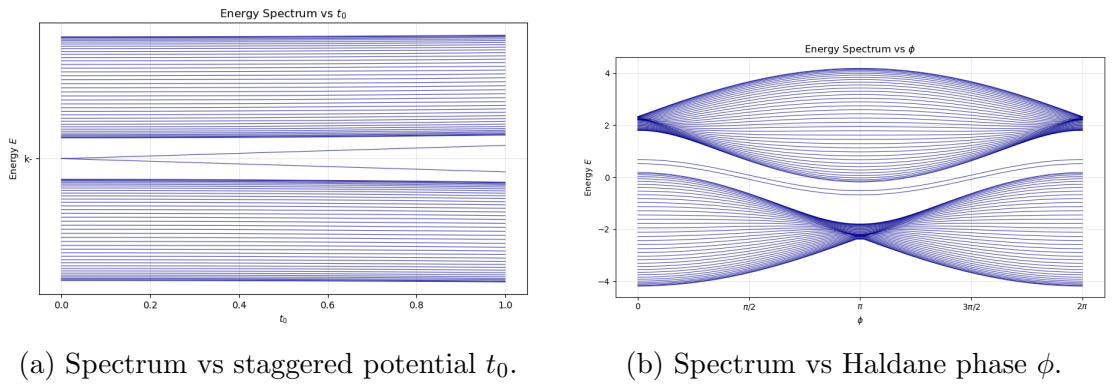


Figure 5.9: Ribbon energy spectrum of the Haldane model as a function of (a) t_0 and (b) ϕ . The edge states (colored lines) merge into the bulk at each gap closing, signaling topological phase transitions.

momentum in both directions, so the entire band structure rotates periodically in momentum space.

The circular drive has two key advantages over the linear drive used in 1D:

1. It *dynamically* breaks time-reversal symmetry by imposing a preferred sense of rotation, which is the key ingredient needed to generate nonzero Chern numbers (Section 1.6.2).
2. On the honeycomb lattice, it can address the K and K' valleys with *different* instantaneous field strengths at each moment, enabling valley-selective gap closing — a capability entirely absent for linearly polarized driving.

Table 5.1 summarizes the key differences between the 1D and 2D driven systems studied in this thesis.

Table 5.1: Comparison of 1D (Chapter 4) and 2D (this chapter) driven systems.

Property	1D (SSH / Rice-Mele)	2D (QWZ / Haldane)
Topological invariant	Winding ν / Zak phase γ	Chern number $\mathcal{C}$
Drive polarization	Linear	Circular
Current	Scalar $J(t)$	Vector (J_x, J_y)
Primary HHG signal	Even-to-odd ratio	Circular dichroism
Static Chern classes	$\nu = 0, 1$	$\mathcal{C} = 0, \pm 1$

5.3.2 The Periodically Driven QWZ Model

Under the circular drive the QWZ Hamiltonian becomes time-periodic:

$$H_{\text{QWZ}}(\mathbf{k}, t) = \sin(k_x + A_x) \sigma_x + \sin(k_y + A_y) \sigma_y + [t_0 + \cos(k_x + A_x) + \cos(k_y + A_y)] \sigma_z. \quad (5.9)$$

The instantaneous energy bands $E_{\pm}(\mathbf{k}, t) = \pm |\mathbf{d}(\mathbf{k}, t)|$ rotate continuously in momentum space as $\mathbf{A}(t)$ sweeps its circle. Figure 5.10 tracks how the conduction band and high-symmetry-path spectrum evolve with increasing $|\mathbf{A}|$.

5.3.3 The Periodically Driven Haldane Model

The circular drive on the honeycomb lattice introduces a richer phenomenology because the two Dirac valleys respond differently at each instant. The time-periodic Haldane Hamiltonian is:

$$H_{\text{H}}(\mathbf{k}, t) = \begin{pmatrix} d_z(\mathbf{k}, t) & f(\mathbf{k}, t) \\ f^*(\mathbf{k}, t) & -d_z(\mathbf{k}, t) \end{pmatrix}, \quad (5.10)$$

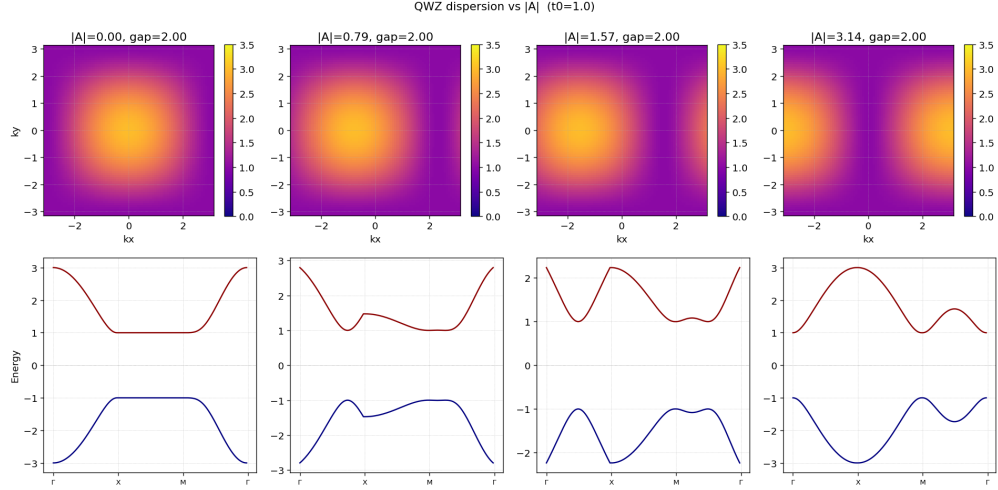


Figure 5.10: QWZ model ($t_0 = 1.0$) under circular drive. **Top:** Conduction band $E_+(\mathbf{k})$ for increasing $|A|$. **Bottom:** High-symmetry-path spectrum. The gap minimum migrates through the Brillouin zone and closes at critical $|A|$, signaling a topological transition.

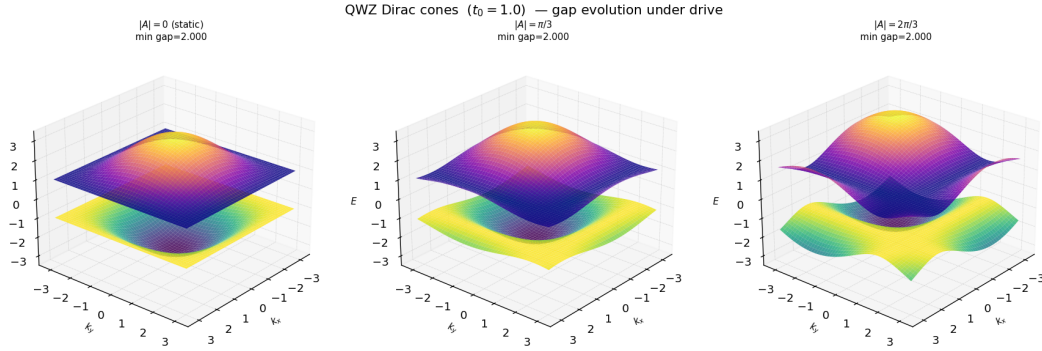


Figure 5.11: Evolution of the Dirac cone structure in the QWZ model as $|A|$ increases. At critical values, the cone gap closes and reopens with a different winding, marking light-induced topological transitions.

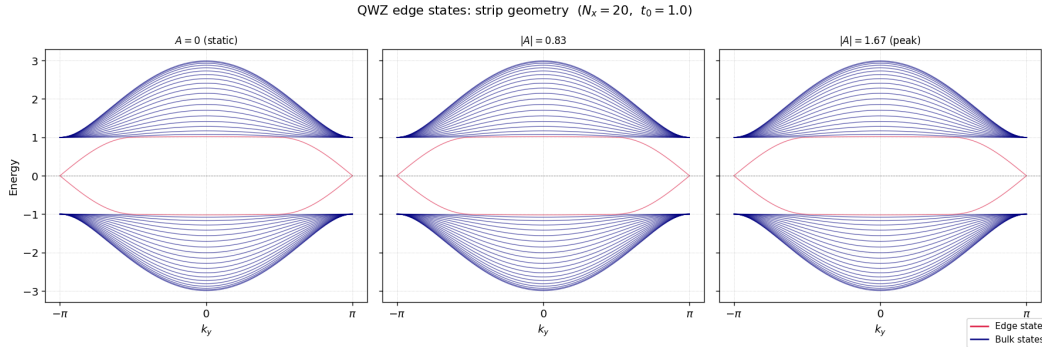


Figure 5.12: Edge states of the QWZ ribbon ($N_x = 20$, $t_0 = 1.0$) under increasing circular drive. Red lines: edge states. Navy: bulk states. The chiral edge modes persist under moderate driving; they annihilate at the transition.

where $f(\mathbf{k}, t) = -t_1 \sum_j e^{i(\mathbf{k} + \mathbf{A}(t)) \cdot \mathbf{a}_j}$ and $d_z(\mathbf{k}, t) = M - 2t_2 \sin \phi \sum_j \sin((\mathbf{k} + \mathbf{A}(t)) \cdot \mathbf{b}_j)$.

The effective masses at the Dirac points under the drive are $d_z(K, t)$ and $d_z(K', t)$, which oscillate out of phase with each other (Fig. ??). This **valley-selective** response is the hallmark of the honeycomb geometry under circular driving: right-circular light can preferentially close the gap at K while leaving K' gapped, or vice versa, by tuning $|\mathbf{A}|$ to the appropriate critical value. This asymmetry is inaccessible to linearly polarized light and is the key mechanism by which Floquet engineering can invert the Chern number of a single valley without affecting the other.

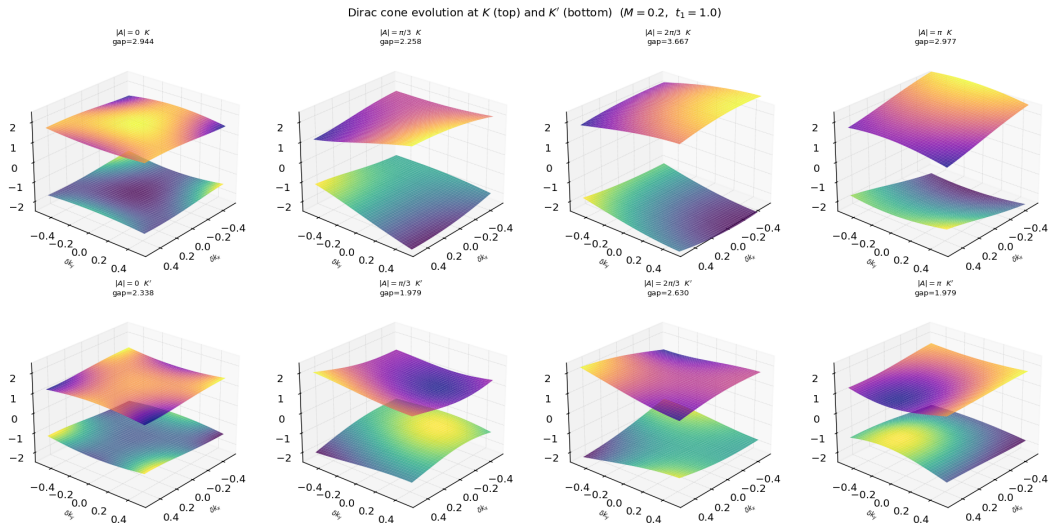


Figure 5.13: Dirac cone evolution at the K (top) and K' (bottom) valleys of the Haldane model ($M = 0.2$, $t_1 = 1.0$) under increasing $|\mathbf{A}|$. The circular drive distorts the cones asymmetrically, closing the gap at each valley at different critical amplitudes.

5.4 Floquet Topological Phase Transitions

5.4.1 Floquet Quasi-Energy Spectra

The full Floquet characterization requires computing the one-cycle propagator $U_F(\mathbf{k}) = \mathcal{T} \exp(-i \int_0^T H(\mathbf{k}, t) dt)$ over the Brillouin zone. The quasi-energies are extracted from the eigenvalues of $U_F(\mathbf{k})$ and confined to the Floquet-Brillouin zone $[-\pi/T, \pi/T)$. The Floquet Chern number C_F is computed via the Fukui-Hatsugai-Suzuki lattice method, which constructs a link variable from overlaps of adjacent Floquet modes and sums the resulting curvature plaquettes over the Brillouin zone [48].

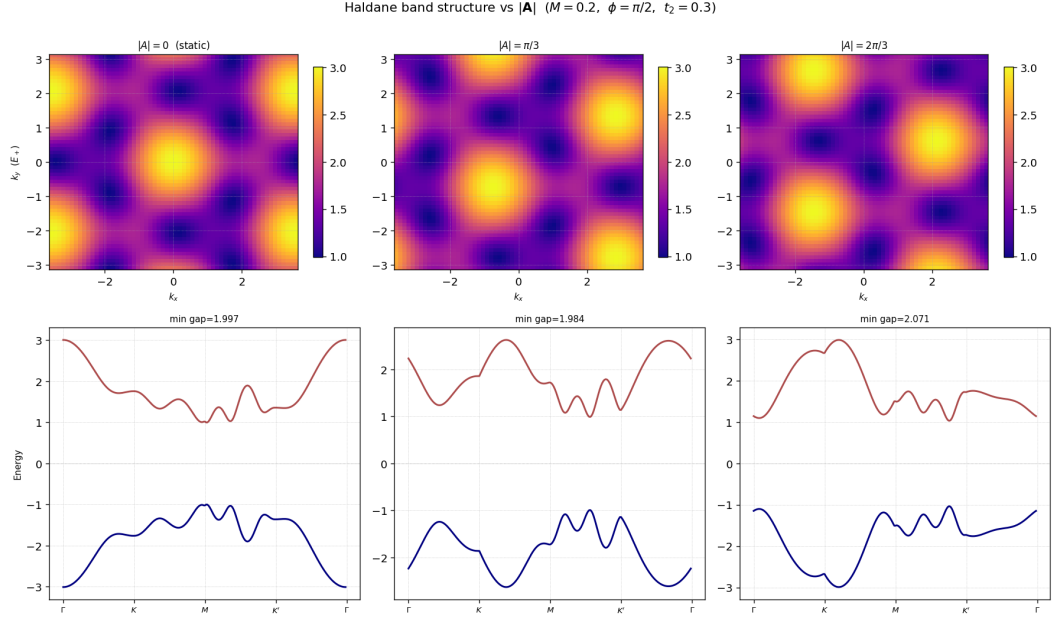


Figure 5.14: Haldane band structure ($M = 0.2$, $\phi = \pi/2$, $t_2 = 0.3$) under circular drive. **Top:** Conduction band in the 2D Brillouin zone. **Bottom:** Path along $\Gamma \rightarrow K \rightarrow M \rightarrow K' \rightarrow \Gamma$. The inequivalent evolution at K and K' is evident.

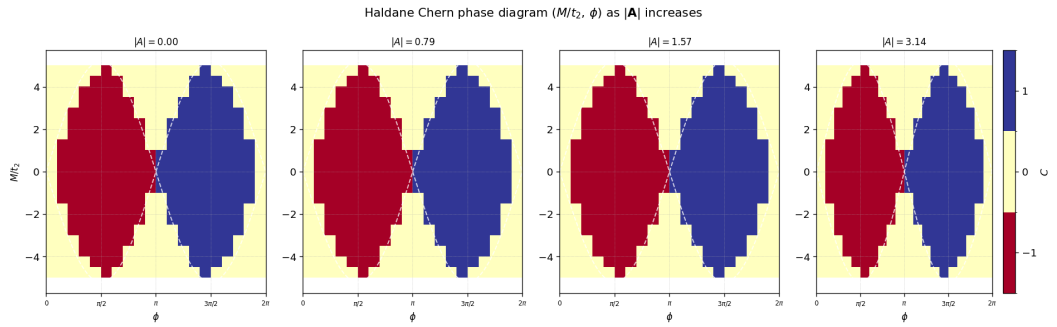


Figure 5.15: Chern number phase diagram of the Haldane model in the $(M/t_2, \phi)$ plane for increasing $|\mathbf{A}|$. The static boundaries (white dashed lines: $M = \pm 3\sqrt{3}t_2 \sin \phi$) shift and deform under driving, and new topological phases with $|C| > 1$ emerge at large amplitudes.

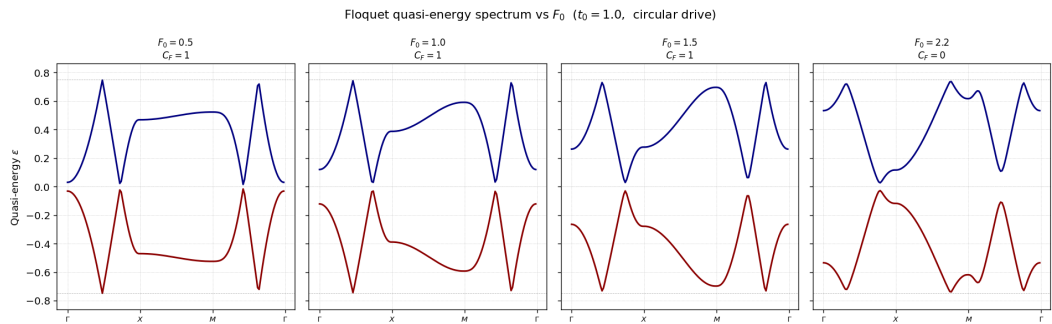


Figure 5.16: Floquet quasi-energy spectrum of the QWZ model ($t_0 = 1.0$) along $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$ for representative drive amplitudes F_0 . The Floquet Chern number C_F is shown in each panel. Gap closings at $\varepsilon = 0$ and $\varepsilon = \pm\pi/T$ mark topological transitions.

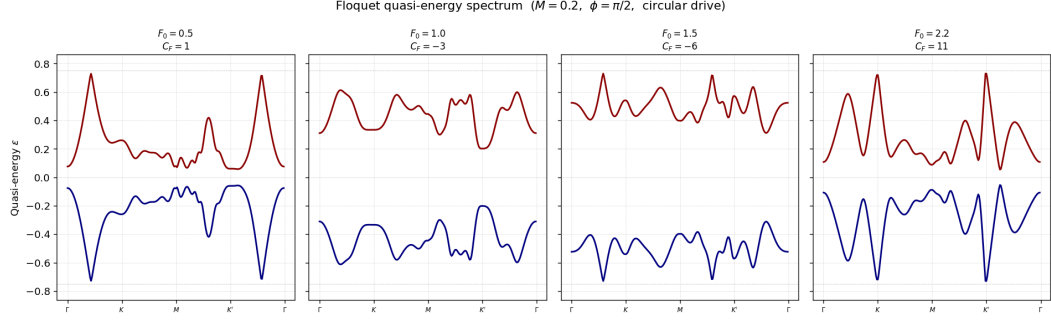


Figure 5.17: Floquet quasi-energy spectrum of the Haldane model ($M = 0.2$, $\phi = \pi/2$) along $\Gamma \rightarrow K \rightarrow M \rightarrow K' \rightarrow \Gamma$. The valley-asymmetric gap closing produces distinct quasi-energy dispersions near K and K' , visible already at moderate F_0 .

5.4.2 Floquet Chern Numbers and Phase Diagrams

Figure 5.18 tracks $C_F(F_0)$ for three representative static configurations of the QWZ model. Each step-like change in C_F is tied to a gap closing in the Floquet spectrum. At larger F_0 , the drive can push the system through multiple transitions, generating Floquet phases with $|C_F| > 1$ that have no static counterpart.

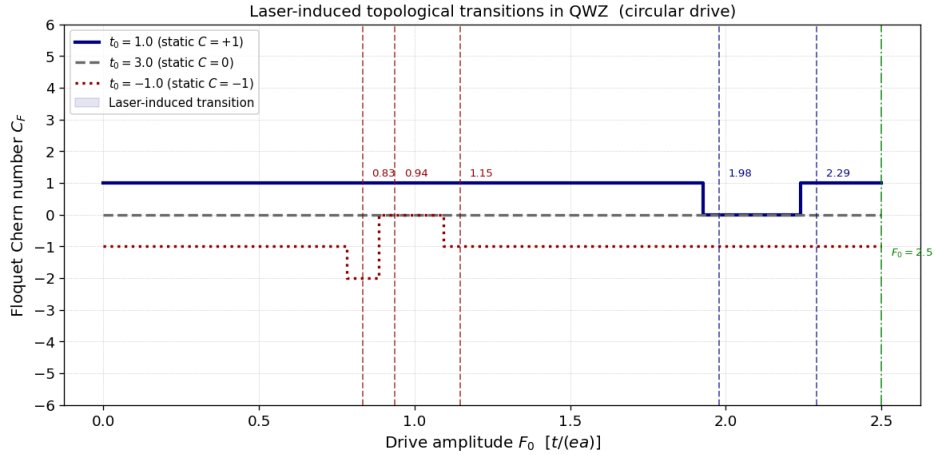


Figure 5.18: Floquet Chern number C_F vs drive amplitude F_0 for three QWZ configurations: $t_0 = 1.0$ ($C = +1$, navy), $t_0 = 3.0$ (trivial, gray), $t_0 = -1.0$ ($C = -1$, red). Step-like transitions occur at critical values F_0^* . The green dashed line marks the working amplitude $F_0 = 2.5$.

For the Haldane model, the valley-selective gap closing produces a distinctive $C_F(F_0)$ curve (Fig. 5.19) in which the two valleys contribute their Chern number changes at different F_0^* values. The transverse Hall conductance $\sigma_{xy}(t) \propto \int_{\text{BZ}} \Omega_z(\mathbf{k}, t) d^2k$ is tracked over the pulse duration in Fig. ??, showing that it oscillates rapidly during the drive and saturates to a quantized value determined by C_F after the pulse.

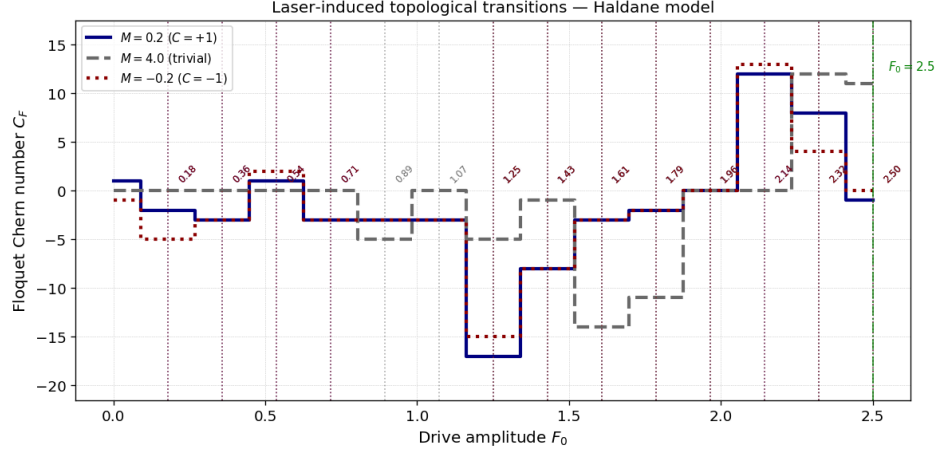


Figure 5.19: C_F vs F_0 for the Haldane model: $M = +0.2$ ($C = +1$, navy), $M = -0.2$ ($C = -1$, red), and $M = 4.0$ (trivial, gray). The two valleys contribute their transitions at different critical amplitudes, producing a staircase pattern with intermediate plateaus.

5.5 High Harmonic Generation in 2D Systems

5.5.1 Observables for 2D HHG Under Circular Drive

In 2D, the macroscopic current is a vector $(J_x(t), J_y(t))$, with each component computed as the expectation value of the corresponding Peierls-derived current operator (Section 2.2.3):

$$J_\mu(t) = \langle \Psi(\mathbf{k}, t) | \hat{J}_\mu(t) | \Psi(\mathbf{k}, t) \rangle, \quad \hat{J}_\mu(t) = -\frac{\partial H(\mathbf{k}, \mathbf{A}(t))}{\partial A_\mu}, \quad \mu \in \{x, y\}. \quad (5.11)$$

The time evolution is computed by integrating the time-dependent Schrödinger equation with a 4th-order Runge-Kutta scheme, initializing in the filled ground state. The HHG power spectrum of each component is:

$$S_\mu(\Omega) = \left| \int_0^{t_{\max}} J_\mu(t) e^{i\Omega t} dt \right|^2. \quad (5.12)$$

The helicity-resolved spectra are then obtained by projecting onto the circular basis $J_\pm = J_x \pm iJ_y$, which separates left- and right-circularly polarized harmonic emission.

For circularly polarized driving, it is natural and physically revealing to project the current onto circular helicity components:

$$J_+(t) = J_x(t) + iJ_y(t) \quad (\text{co-rotating}), \quad J_-(t) = J_x(t) - iJ_y(t) \quad (\text{counter-rotating}). \quad (5.13)$$

The photon angular momentum selection rules govern which harmonics appear in J_+ vs J_- . These rules are dictated by the point-group symmetry of the lattice (Section 3.5.1):

- **Square lattice (QWZ):** C_4 symmetry implies odd harmonics in J_+ , even in J_- (for right-circular drive).
- **Honeycomb lattice (Haldane):** C_3 symmetry implies $3n+1$ harmonics in J_+ , $3n+2$ harmonics in J_- , and harmonics of order $3n$ are suppressed.

Departures from these selection rules are direct spectroscopic indicators of symmetry breaking by the Floquet drive.

5.5.2 HHG Signatures in the QWZ Model

Figure 5.20 shows the total HHG spectrum $S_{\text{tot}} = |J_x|^2 + |J_y|^2$ for the QWZ model in the topological and trivial phases. Unlike the 1D case, the circular drive already breaks the half-cycle symmetry, so both even and odd harmonics are present in both phases. The distinguishing signal is therefore not the presence of even harmonics but rather their *relative yield* compared to the odd ones, which is sensitive to the Chern number.

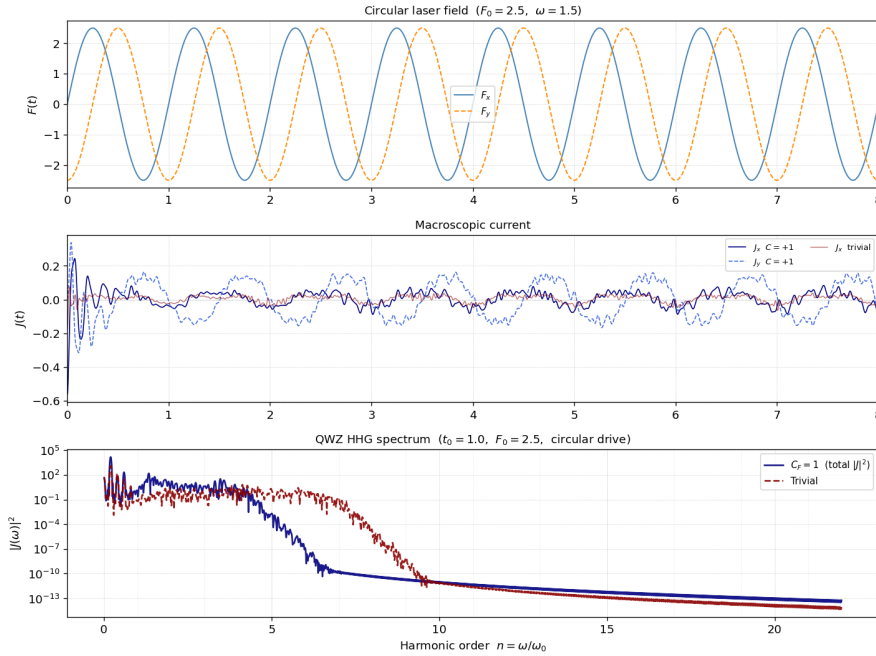


Figure 5.20: Total HHG spectrum $|J|^2$ of the QWZ model: topological phase ($t_0 = 1.0, C = +1$, navy) and trivial phase ($t_0 = 3.0$, red) under circular drive. The current time traces are shown in the middle panel. Both even and odd harmonics are present due to the broken half-cycle symmetry of the circular drive.

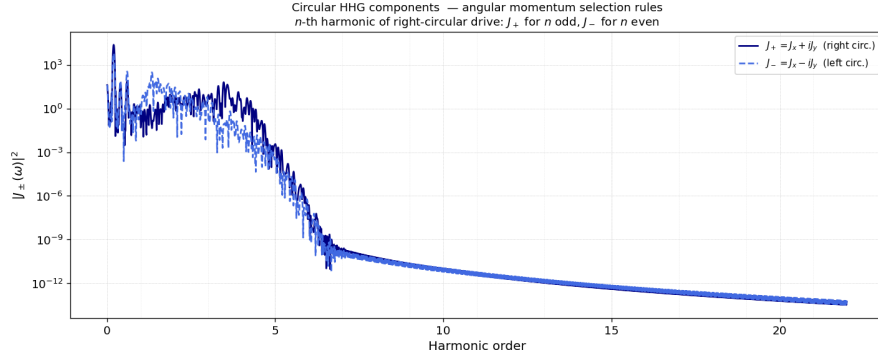


Figure 5.21: Helicity decomposition of the QWZ HHG spectrum. Co-rotating (J_+ , navy) and counter-rotating (J_- , blue) components. Consistent with C_4 symmetry and right-circular driving: odd harmonics dominate J_+ , even harmonics dominate J_- .

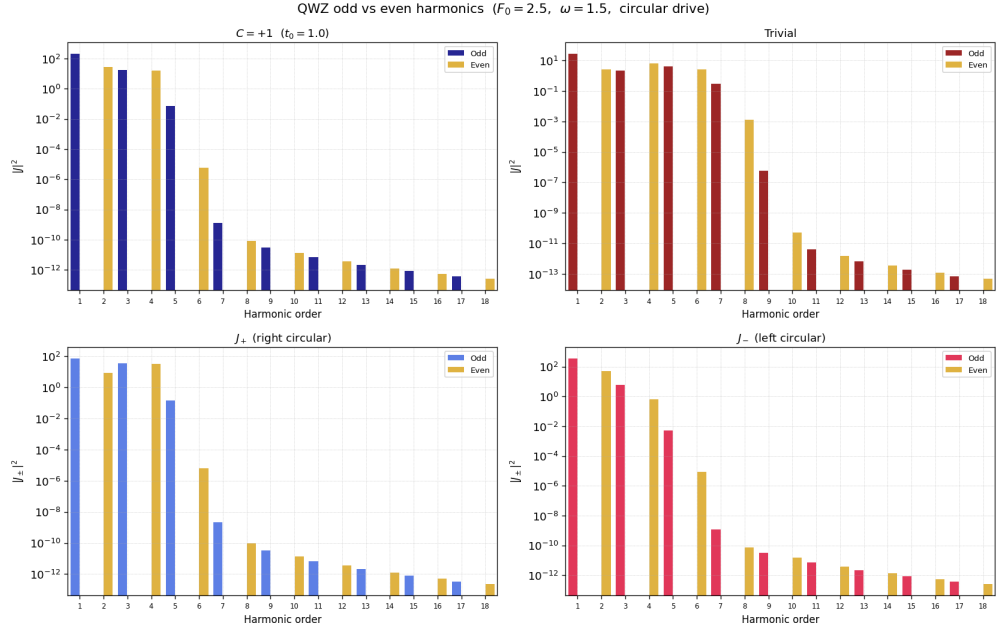


Figure 5.22: Odd/even harmonic decomposition of the QWZ model under circular drive ($F_0 = 2.5$, $\omega = 1.5$). The topological phase (left column) shows a distinct harmonic yield pattern relative to the trivial phase (right column), reflecting the different Floquet Chern numbers.

5.5.3 HHG Signatures in the Haldane Model

The C_3 symmetry of the honeycomb lattice modifies the selection rules. Harmonics of order $3n$ are forbidden; those of order $3n+1$ appear in J_+ and those of order $3n+2$ appear in J_- . The physical origin is the three-fold rotational invariance of the honeycomb lattice combined with the angular momentum of the circularly polarized photons.

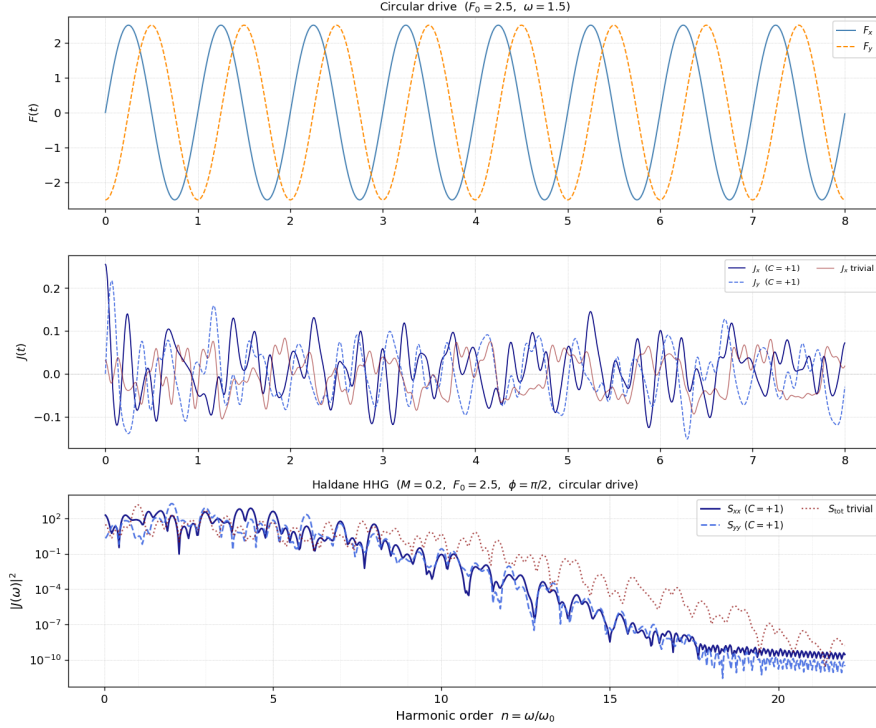


Figure 5.23: HHG spectrum of the Haldane model ($M = 0.2$, $C = +1$, $\phi = \pi/2$) under circular drive, compared to the trivial phase (red). The C_3 symmetry suppresses harmonics at orders $3, 6, 9, \dots$. The longitudinal (S_{xx} , navy) and transverse (S_{yy} , blue) components are shown separately.

5.5.4 Summary of 2D Topological HHG Signatures

The results of this chapter establish four clear topological fingerprints in the 2D HHG spectra:

1. **Harmonic yield modulation:** The total harmonic yield and the relative intensities of different harmonic orders differ systematically between phases with different Chern numbers, reflecting the distinct Berry curvature distributions of the Floquet bands.
2. **Helicity selection rules:** The C_4 symmetry of the square lattice (QWZ) and the C_3 symmetry of the honeycomb lattice (Haldane) each impose spe-

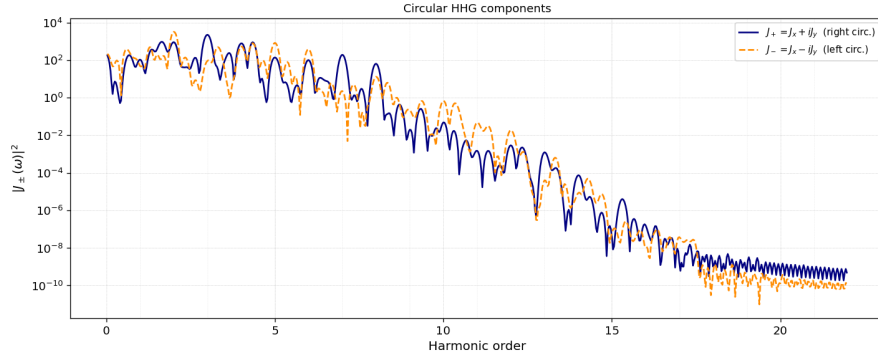


Figure 5.24: Helicity decomposition of the Haldane HHG spectrum. J_+ (navy) carries harmonics $3n+1$ (1, 4, 7, ...); J_- (orange) carries harmonics $3n+2$ (2, 5, 8, ...). Harmonics $3n$ are absent, confirming the C_3 selection rule.

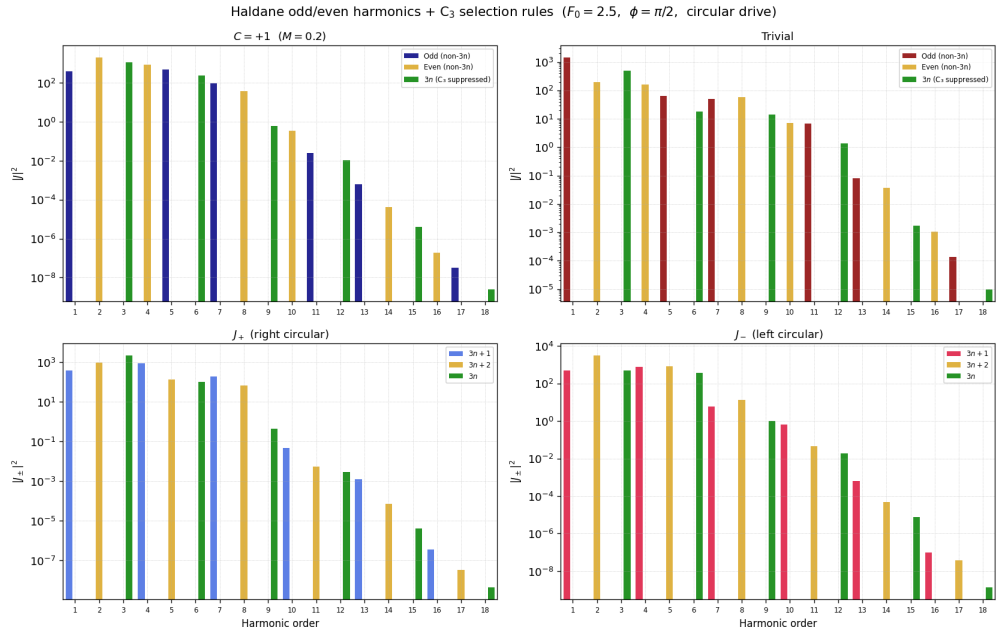


Figure 5.25: Full harmonic decomposition of the Haldane model with C_3 selection rules. **Top row:** Total spectrum for topological (left) and trivial (right) phases, with $3n$ -suppressed harmonics highlighted in green. **Bottom row:** J_+ (left) and J_- (right) components, decomposed into $3n+1$, $3n+2$, and $3n$ groups.

cific angular momentum selection rules on the co- and counter-rotating current components. These rules are modified when the Floquet drive lowers the effective symmetry.

3. **Valley-selective emission in the Haldane model:** Because the circular drive addresses the K and K' valleys asymmetrically, it produces valley-polarized harmonic emission whose polarization and intensity encode the valley-resolved Berry curvature.
4. **All-optical detection of Floquet transitions:** As F_0 crosses the critical amplitudes F_0^* where C_F changes (Figs. 5.18 and 5.19), the HHG spectra undergo abrupt qualitative changes. These changes provide an ultrafast, all-optical clock for the onset of light-induced topological order.

Taken together, these signatures demonstrate that HHG spectroscopy — particularly the helicity-resolved harmonic spectrum under circularly polarized driving — is a powerful probe of 2D topological phases that is both symmetry-sensitive and capable of resolving non-equilibrium Floquet transitions in real time.

Conclusion

This thesis developed a unified framework connecting topological band theory, Floquet engineering, and solid-state HHG, applied to four paradigmatic models in one and two dimensions.

Topological invariants as geometric quantities. The distinction between trivial and topological insulators is encoded in the geometry of occupied Bloch eigenstates over the Brillouin zone. The Berry curvature acts as a fictitious magnetic field in momentum space; its integral yields the Zak phase (1D) or Chern number (2D), which can change only when the bulk gap closes. The bulk–boundary correspondence then guarantees protected gapless boundary states.

Floquet engineering of topological phases. Periodic laser driving induces topological phase transitions with no equilibrium counterpart. In the SSH model, the Floquet winding number $\nu_F \in \{0, 1, 2\}$ is tunable via drive amplitude F_0 , with phase boundaries set by quasienergy gap closings at $\varepsilon = 0$ or $\pm\pi/T$. The same phenomenology holds for the Rice-Mele model with the Floquet Zak phase. Under circularly polarized driving, the Floquet Chern number of the QWZ and Haldane models passes through multiple integers, including $|C_F| > 1$, inaccessible to any static Hamiltonian. In the Haldane model, the drive closes the gap selectively at K or K' , enabling valley-resolved Floquet engineering.

HHG as an all-optical topology probe. The topological invariant leaves clear signatures in the harmonic spectrum. In 1D, the even-to-odd harmonic ratio is substantially larger in the topological phase, owing to edge-state contributions that break half-cycle symmetry locally. $R(F_0)$ tracks Floquet transitions with high fidelity, making HHG a real-time sensor of non-equilibrium topological order. In 2D, helicity-resolved spectra encode both crystal symmetry and Chern number: C_4 symmetry in QWZ places odd harmonics in J_+ and even in J_- ; C_3 symmetry in Haldane places $3n+1$ harmonics in J_+ , $3n+2$ in J_- , and suppresses $3n$ harmonics entirely.

Robustness. In the Rice-Mele model, the staggered potential Δ breaks chiral symmetry and shifts edge states away from zero energy, yet the enhanced even-harmonic yield persists as long as inversion symmetry is maintained, confirming R as a reliable probe for a class of inversion-symmetric 1D topological insulators.

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Ethics and Risk Management Declaration

Ethics and Professional Conduct

This thesis investigates light-induced topological phase transitions in 1D and 2D topological insulator models using numerical simulation. The work involved the independent development of Python codes to compute band structures, Berry phases, Chern numbers, Floquet quasi-energy spectra, and HHG power spectra for the SSH, Rice-Mele, QWZ, and Haldane models. The following ethical principles governed every stage of the project.

1. **Academic integrity and originality.** All numerical implementations — including the tight-binding Hamiltonian constructors, the 4th-order Runge-Kutta time propagators, the Fukui-Hatsugai-Suzuki Chern number algorithm, and the Floquet propagator routines — were written from scratch by the author. Standard open-source libraries (`NumPy`, `SciPy`, `Matplotlib`) were used strictly for numerical operations and visualization. No code was copied from unpublished sources. Every figure in the thesis was generated by the author’s own scripts, and all external results used for validation are explicitly cited.
2. **Honest representation of results.** Topological invariants are computed with specified k -point mesh densities, and convergence of the Chern number and winding number with respect to mesh size was verified in all cases. Where numerical results approach a phase boundary and the invariant is sensitive to small changes in parameters, this is explicitly noted in the text. No results have been selectively omitted or adjusted to match prior expectations.
3. **Proper attribution.** The foundational theoretical results used in this thesis — Berry phase, Chern number, Floquet theorem, HHG decomposition — are well-established in the literature and are cited accordingly. The specific models studied (SSH, Rice-Mele, QWZ, Haldane) are attributed to their original authors. The thesis makes clear which results are reproductions of

known benchmarks (used for validation) and which are original numerical calculations.

4. **Transparency and reproducibility.** All code used to produce the figures and results in this thesis is publicly available at <https://github.com/Hazem-74/Topological-Insulator-Models>. The repository includes the model parameters, laser parameters, and mesh specifications needed to reproduce every result. This ensures that the findings are verifiable and that others can build on this work.
5. **Appropriate use of computational resources.** The simulations were designed to be computationally efficient, running on a standard workstation without requiring dedicated cluster resources. Convergence tests were used to select the minimum mesh density sufficient for accurate results, avoiding unnecessary computation.
6. **Supervision and collaboration.** This project was conducted under the supervision of Dr. Abdallah AlShafey. All physical interpretation and numerical implementation were carried out independently by the author. The supervisor provided guidance on the scope and direction of the project. No results attributed to the author in this thesis were produced by the supervisor or by other collaborators.

Risk Management

The following risks were identified at the outset of the project and managed throughout.

1. **Numerical correctness of topological invariants.**

Risk: The Chern number and winding number are integer-valued but computed by integrating smooth functions over a discrete k -point mesh. A coarse mesh near a phase boundary can yield fractional or incorrect integers.

Mitigation: All topological invariants were computed using gauge-invariant, lattice-formulated methods (the Zak phase via parallel transport, the Chern number via the Fukui-Hatsugai-Suzuki plaquette sum). Convergence with mesh density was verified for each model and parameter set. Results near phase boundaries were flagged and cross-checked by computing the invariant at slightly displaced parameter values on both sides of the transition.

2. **Accuracy of the Floquet propagator.**

Risk: Time-ordering the Floquet propagator numerically requires a sufficiently fine time step; coarse integration can introduce unphysical quasi-energy gaps or miss crossings.

Mitigation: A 4th-order Runge-Kutta integrator was used with a time step equal to $T/500$, where T is the driving period. The unitarity of $U_F(\mathbf{k})$ was checked numerically after each propagation as a diagnostic.

3. Interpretation of HHG spectra near topological transitions.

Risk: Near a Floquet transition, the quasi-energy gap is small and the system is highly susceptible to driving, potentially leading to HHG spectra that are difficult to interpret.

Mitigation: HHG simulations were performed with a smooth pulse envelope (sine-squared) to avoid spectral leakage, and a Hann window was applied before Fourier transformation. Results at phase boundaries were presented as signatures of the transition rather than characteristics of a well-defined phase, consistent with the physical picture of a closing gap.

4. Validation against known benchmarks.

Risk: Without experimental data for direct comparison, it is possible to produce plausible-looking but incorrect results.

Mitigation: Each model was first validated against published analytical and numerical results: the SSH edge states and winding number against [14], the QWZ Chern number phase diagram against [47], the Haldane phase diagram against [46], and the HHG odd-harmonic selection rule against the symmetry argument in Appendix C.2. Only after these benchmarks were reproduced correctly was the Floquet and HHG analysis carried out.

5. Scope and timeline management.

Risk: The project spans four models, three theoretical frameworks, and three levels of analysis (static, Floquet, HHG), creating a risk of scope creep and delayed completion.

Mitigation: A phased approach was adopted: 1D models (SSH, Rice-Mele) were completed before 2D models (QWZ, Haldane), and within each model the static analysis preceded the Floquet and HHG analysis. Regular progress meetings with the supervisor were used to prioritize tasks and adjust the scope as needed. No major milestones were missed.

Appendix A: Derivations for Topological Band Theory

This appendix provides the detailed mathematical derivations for the tight-binding dispersion relation and the geometric Berry phase of a generic two-band model, which were introduced in Chapter 1.

A.1 Derivation of the Tight-Binding Dispersion

In Section 1.3, we introduced the real-space Hamiltonian for a one-dimensional chain of N atoms with lattice spacing a and periodic boundary conditions:

$$H = E_0 \sum_{n=1}^N |n\rangle \langle n| - t \sum_{n=1}^N (|n\rangle \langle n+1| + |n+1\rangle \langle n|), \quad (\text{A.1})$$

where E_0 is the on-site energy, t is the nearest-neighbor hopping amplitude, and $|n\rangle$ represents an electron localized at site n . We impose periodic boundary conditions such that $|N+1\rangle \equiv |1\rangle$.

To find the energy eigenvalues, we construct the Bloch state ansatz. A normalized eigenstate with crystal momentum k in the first Brillouin zone ($k \in [-\pi/a, \pi/a)$) is given by the discrete Fourier transform of the localized states:

$$|\psi_k\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ikna} |n\rangle. \quad (\text{A.2})$$

We now apply the Hamiltonian H to the state $|\psi_k\rangle$:

$$H |\psi_k\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ikna} H |n\rangle. \quad (\text{A.3})$$

Acting with H on the localized state $|n\rangle$ yields:

$$H |n\rangle = E_0 |n\rangle - t |n+1\rangle - t |n-1\rangle. \quad (\text{A.4})$$

Substituting this back into the expression for $H|\psi_k\rangle$, we obtain:

$$H|\psi_k\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ikna} (E_0|n\rangle - t|n+1\rangle - t|n-1\rangle). \quad (\text{A.5})$$

We can separate this into three sums. The first sum is simply $E_0|\psi_k\rangle$. For the second and third sums, we perform a change of summation indices. Let $m = n+1$ in the second sum, and $m = n-1$ in the third sum. Because of the periodic boundary conditions, the limits of summation remain unchanged, yielding:

$$\begin{aligned} H|\psi_k\rangle &= E_0|\psi_k\rangle - \frac{t}{\sqrt{N}} \sum_{m=1}^N e^{ik(m-1)a} |m\rangle - \frac{t}{\sqrt{N}} \sum_{m=1}^N e^{ik(m+1)a} |m\rangle \\ &= E_0|\psi_k\rangle - t(e^{-ika} + e^{ika}) \left(\frac{1}{\sqrt{N}} \sum_{m=1}^N e^{ikma} |m\rangle \right). \end{aligned} \quad (\text{A.6})$$

Recognizing the term in the parentheses as the original Bloch state $|\psi_k\rangle$, we arrive at the eigenvalue equation:

$$H|\psi_k\rangle = [E_0 - t(e^{ika} + e^{-ika})] |\psi_k\rangle. \quad (\text{A.7})$$

Using Euler's formula, $e^{ika} + e^{-ika} = 2\cos(ka)$, the energy eigenvalue (dispersion relation) is:

$$E(k) = E_0 - 2t\cos(ka). \quad (\text{A.8})$$

This confirms the dispersion relation stated in Eq. (1.5).

A.2 Berry Phase and Curvature for a Two-Band Model

In Section 1.4, we considered a generic two-level Hamiltonian expressed in terms of Pauli matrices $\boldsymbol{\sigma}$:

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma} = \begin{pmatrix} d_z & d_x - id_y \\ d_x + id_y & -d_z \end{pmatrix}, \quad (\text{A.9})$$

where $\mathbf{d}(\mathbf{k}) = (d_x, d_y, d_z)$ is a vector in parameter space. The eigenvalues are $E_{\pm}(\mathbf{k}) = \pm|\mathbf{d}(\mathbf{k})|$. The eigenvectors depend only on the direction of the unit vector $\hat{\mathbf{d}}(\mathbf{k}) = \mathbf{d}(\mathbf{k})/|\mathbf{d}(\mathbf{k})|$.

We parameterize $\hat{\mathbf{d}}$ using spherical coordinates (θ, ϕ) :

$$\hat{\mathbf{d}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta). \quad (\text{A.10})$$

The normalized ground state spinor $|u_{-}\rangle$, corresponding to the eigenvalue $-|\mathbf{d}|$, points in the $-\hat{\mathbf{d}}$ direction on the Bloch sphere and is given by:

$$|u_{-}\rangle = \begin{pmatrix} -\sin(\theta/2)e^{-i\phi} \\ \cos(\theta/2) \end{pmatrix}. \quad (\text{A.11})$$

Derivation of the Berry Connection and Phase

The Berry connection is defined as $\mathbf{A} = i \langle u_{-} | \nabla | u_{-} \rangle$. In the spherical parameter space (θ, ϕ) , we compute the components A_{θ} and A_{ϕ} .

First, we calculate the derivatives of the state $|u_{-}\rangle$:

$$\partial_{\theta} |u_{-}\rangle = \begin{pmatrix} -\frac{1}{2} \cos(\theta/2) e^{-i\phi} \\ -\frac{1}{2} \sin(\theta/2) \end{pmatrix}, \quad \partial_{\phi} |u_{-}\rangle = \begin{pmatrix} i \sin(\theta/2) e^{-i\phi} \\ 0 \end{pmatrix}. \quad (\text{A.12})$$

Now, we evaluate the inner products:

$$\langle u_{-} | \partial_{\theta} | u_{-} \rangle = (-\sin(\theta/2) e^{i\phi}) \left(-\frac{1}{2} \cos(\theta/2) e^{-i\phi} \right) + (\cos(\theta/2)) \left(-\frac{1}{2} \sin(\theta/2) \right) = 0, \quad (\text{A.13})$$

$$\langle u_{-} | \partial_{\phi} | u_{-} \rangle = (-\sin(\theta/2) e^{i\phi}) (i \sin(\theta/2) e^{-i\phi}) + 0 = -i \sin^2(\theta/2). \quad (\text{A.14})$$

Thus, the components of the Berry connection are:

$$A_{\theta} = i \langle u_{-} | \partial_{\theta} | u_{-} \rangle = 0, \quad A_{\phi} = i \langle u_{-} | \partial_{\phi} | u_{-} \rangle = \sin^2(\theta/2) = \frac{1 - \cos \theta}{2}. \quad (\text{A.15})$$

Consider a closed loop C at a constant polar angle θ , where the azimuthal angle ϕ traverses from 0 to 2π . The Berry phase γ_C acquired along this loop is:

$$\gamma_C = \oint_C \mathbf{A} \cdot d\mathbf{l} = \int_0^{2\pi} A_{\phi} d\phi = 2\pi \left(\frac{1 - \cos \theta}{2} \right) = \pi(1 - \cos \theta). \quad (\text{A.16})$$

The solid angle Ω_{solid} subtended by this loop at the origin of the Bloch sphere is the area of the spherical cap:

$$\Omega_{\text{solid}} = \int_0^{2\pi} d\phi \int_0^{\theta} \sin \theta' d\theta' = 2\pi(1 - \cos \theta). \quad (\text{A.17})$$

Comparing Eq. (A.16) with the solid angle, we immediately obtain the geometric relation stated in Eq. (1.11):

$$\gamma_C = \frac{1}{2} \Omega_{\text{solid}}. \quad (\text{A.18})$$

Derivation of the Berry Curvature

The Berry curvature in the parameter space (θ, ϕ) is the "magnetic field" derived from the connection, given by the antisymmetric tensor component $F_{\theta\phi}$:

$$F_{\theta\phi} = \partial_\theta A_\phi - \partial_\phi A_\theta = \partial_\theta \left(\frac{1 - \cos \theta}{2} \right) - 0 = \frac{1}{2} \sin \theta. \quad (\text{A.19})$$

The infinitesimal Berry phase flux through an area element $d\theta d\phi$ is:

$$d\gamma = F_{\theta\phi} d\theta d\phi = \frac{1}{2} \sin \theta d\theta d\phi. \quad (\text{A.20})$$

Recognizing that the infinitesimal solid angle element on the sphere is $d\Omega_{\text{solid}} = \sin \theta d\theta d\phi$, we have $d\gamma = \frac{1}{2} d\Omega_{\text{solid}}$.

To express this in the original momentum space $\mathbf{k} = (k_x, k_y)$, we note that the mapping $\mathbf{k} \mapsto \hat{\mathbf{d}}(\mathbf{k})$ induces a Jacobian transformation. The scalar Berry curvature F (which corresponds to the z -component Ω_{xy} in 2D) is given by the solid angle element per unit area in $\mathbf{k}$ -space:

$$F = \frac{1}{2} \epsilon_{ij} \hat{\mathbf{d}} \cdot (\partial_{k_i} \hat{\mathbf{d}} \times \partial_{k_j} \hat{\mathbf{d}}), \quad (\text{A.21})$$

where $i, j \in \{x, y\}$ and ϵ_{ij} is the 2D Levi-Civita symbol. This confirms Eq. (1.12) and demonstrates that the Berry curvature measures the density of the solid angle swept by $\hat{\mathbf{d}}(\mathbf{k})$ as $\mathbf{k}$ varies across the Brillouin zone.

Appendix B: Derivations and Formulations for Floquet Theory

This appendix provides the detailed mathematical derivations and formal expressions for the concepts introduced in Chapter 2, including the Peierls substitution, Floquet's theorem, and the high-frequency expansion.

B.1 Derivation of the Peierls Substitution

In Section 2.2, we introduced the Peierls substitution to incorporate a vector potential into a tight-binding Hamiltonian. Here, we derive this result from the minimal coupling prescription.

For a free particle of charge q in a vector potential $\mathbf{A}$, the Hamiltonian is $H = (\hat{\mathbf{p}} - q\mathbf{A})^2/2m$. If we assume $\mathbf{A}$ is spatially constant over the scale of the lattice, we can write the shifted momentum operator as a unitary transformation of the canonical momentum:

$$\hat{\mathbf{p}} - q\mathbf{A} = e^{iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar} \hat{\mathbf{p}} e^{-iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar}. \quad (\text{B.1})$$

This suggests that the effect of the vector potential can be introduced into any Hamiltonian via the unitary gauge transformation $H_{\mathbf{A}} = e^{iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar} H_{\mathbf{A}=0} e^{-iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar}$.

Applying this transformation to the tight-binding hopping term $t_{\mathbf{mn}} |\mathbf{m}\rangle \langle \mathbf{n}|$, where $|\mathbf{m}\rangle$ is a localized Wannier state at position $\mathbf{r}_{\mathbf{m}}$, we get:

$$\begin{aligned} e^{iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar} (t_{\mathbf{mn}} |\mathbf{m}\rangle \langle \mathbf{n}|) e^{-iq\mathbf{A}\cdot\hat{\mathbf{r}}/\hbar} &= t_{\mathbf{mn}} (e^{iq\mathbf{A}\cdot\mathbf{r}_{\mathbf{m}}/\hbar} |\mathbf{m}\rangle) (\langle \mathbf{n}| e^{-iq\mathbf{A}\cdot\mathbf{r}_{\mathbf{n}}/\hbar}) \\ &= t_{\mathbf{mn}} e^{iq\mathbf{A}\cdot(\mathbf{r}_{\mathbf{m}}-\mathbf{r}_{\mathbf{n}})/\hbar} |\mathbf{m}\rangle \langle \mathbf{n}|. \end{aligned} \quad (\text{B.2})$$

Defining the hopping vector $\boldsymbol{\delta} = \mathbf{r}_{\mathbf{m}} - \mathbf{r}_{\mathbf{n}}$, the phase factor is $e^{iq\mathbf{A}\cdot\boldsymbol{\delta}/\hbar}$. For a general, spatially varying vector potential $\mathbf{A}(\mathbf{r}, t)$, the straight-line path integral replaces the dot product, yielding the exact Peierls substitution:

$$t_{\mathbf{mn}} \rightarrow t_{\mathbf{mn}} \exp \left(-\frac{iq}{\hbar} \int_{\mathbf{r}_{\mathbf{n}}}^{\mathbf{r}_{\mathbf{m}}} d\mathbf{r} \cdot \mathbf{A}(\mathbf{r}, t) \right). \quad (\text{B.3})$$

This confirms Eq. (2.2).

B.2 Derivation of the Current Operator Relation

In Section 2.2.3, we defined the current operator as $\hat{\mathbf{J}} = -\partial\hat{H}/\partial\mathbf{A}$. Here, we briefly justify this fundamental relation from continuum, thermodynamic, and lattice perspectives.

Continuum limit. For a particle of charge q and mass m , the Hamiltonian is $\hat{H} = (\hat{\mathbf{p}} - q\mathbf{A})^2/2m + V(\mathbf{r})$. The velocity operator is $\hat{\mathbf{v}} = (\hat{\mathbf{p}} - q\mathbf{A})/m$. Taking the derivative of $\hat{H}$ with respect to $\mathbf{A}$ yields $\partial\hat{H}/\partial\mathbf{A} = -q(\hat{\mathbf{p}} - q\mathbf{A})/m = -q\hat{\mathbf{v}} = -\hat{\mathbf{J}}$.

Thermodynamic conjugate variables. The power delivered by the electromagnetic field is $P = \int \mathbf{J} \cdot \mathbf{E} d^3r$. In the Coulomb gauge ($\mathbf{E} = -\partial_t\mathbf{A}$), this becomes $P = -\int \mathbf{J} \cdot \dot{\mathbf{A}} d^3r$. Equating this to the Hamiltonian rate of change $P = \int (\partial\hat{H}/\partial\mathbf{A}) \cdot \dot{\mathbf{A}} d^3r$ for an arbitrary time-varying vector potential $\dot{\mathbf{A}}$ immediately requires $\hat{\mathbf{J}} = -\partial\hat{H}/\partial\mathbf{A}$.

Tight-binding lattice. For a hopping term modified by the Peierls substitution, $\hat{H}_{ij} = t_{ij} \exp(-i\frac{q}{\hbar}\mathbf{A} \cdot \boldsymbol{\delta}_{ij}) |i\rangle \langle j| + \text{h.c.}$, taking the derivative with respect to $\mathbf{A}$ directly pulls down the displacement vector $\boldsymbol{\delta}_{ij}$:

$$\hat{\mathbf{J}}_{ij} = -\frac{\partial\hat{H}_{ij}}{\partial\mathbf{A}} = i\frac{q}{\hbar}\boldsymbol{\delta}_{ij} t_{ij} \exp\left(-i\frac{q}{\hbar}\mathbf{A} \cdot \boldsymbol{\delta}_{ij}\right) |i\rangle \langle j| + \text{h.c.} \quad (\text{B.4})$$

This confirms that the operator definition correctly captures the physical charge displacement $\boldsymbol{\delta}_{ij}$ during a quantum hop, which is the form utilized in our numerical calculations.

B.3 Derivation of Floquet's Theorem and Time-Evolution Operator

In Section 2.3, we stated Floquet's theorem. Here, we derive it from the fundamental properties of the time-evolution operator. The time-evolution operator is defined as:

$$U(t, t_0) = \mathcal{T} \exp\left(-\frac{i}{\hbar} \int_{t_0}^t dt' H(t')\right). \quad (\text{B.5})$$

The derivation begins with the semigroup property of the time-evolution operator:

$$U(t_1 + t_2, t_0) = U(t_1 + t_2, t_1)U(t_1, t_0). \quad (\text{B.6})$$

For a periodically modulated Hamiltonian $H(t+T) = H(t)$, we consider evolution from an initial time t_0 to a final time $t+T$:

$$\begin{aligned} U(t+T, t_0) &= U(t+T, t_0+T)U(t_0+T, t_0) \\ &= U(t, t_0)U(t_0+T, t_0), \end{aligned} \quad (\text{B.7})$$

where we used the periodicity to establish $U(t+T, t_0+T) = U(t, t_0)$.

Since $U(t_0+T, t_0)$ is a unitary operator, we can define a Hermitian operator $H_F[t_0]$ — the Floquet Hamiltonian — via:

$$U(t_0+T, t_0) \equiv e^{-iH_F[t_0]T/\hbar}. \quad (\text{B.8})$$

We then define the kick operator $\hat{K}_F[t_0](t)$ as:

$$e^{-i\hat{K}_F[t_0](t)} \equiv U(t, t_0)e^{+iH_F[t_0](t-t_0)/\hbar}. \quad (\text{B.9})$$

Using Eq. (B.7), we can verify the periodicity of the kick operator:

$$\begin{aligned} e^{-i\hat{K}_F[t_0](t+T)} &= U(t+T, t_0)e^{+iH_F[t_0](t+T-t_0)/\hbar} \\ &= U(t, t_0) \left(U(t_0+T, t_0)e^{+iH_F[t_0]T/\hbar} \right) e^{+iH_F[t_0](t-t_0)/\hbar} \\ &= U(t, t_0)e^{+iH_F[t_0](t-t_0)/\hbar} = e^{-i\hat{K}_F[t_0](t)}. \end{aligned} \quad (\text{B.10})$$

Rearranging the definition of the kick operator immediately yields Floquet's theorem:

$$U(t, t_0) = e^{-i\hat{K}_F[t_0](t)} e^{-iH_F[t_0](t-t_0)/\hbar}. \quad (\text{B.11})$$

B.4 Derivation of the Sambe Space and the Floquet Operator

We begin with the time-dependent Schrödinger equation for a spatially and temporally periodic Hamiltonian:

$$\left(H(\mathbf{r}, t) - i\hbar \frac{\partial}{\partial t} \right) |\Psi(\mathbf{r}, t)\rangle = 0. \quad (\text{B.12})$$

Substituting the Floquet-Bloch ansatz $|\Psi_{\alpha, \mathbf{k}}(\mathbf{r}, t)\rangle = e^{i\mathbf{k} \cdot \mathbf{r} - i\varepsilon_{\alpha, \mathbf{k}} t/\hbar} |u_{\alpha, \mathbf{k}}(\mathbf{r}, t)\rangle$ yields the eigenvalue equation for the Floquet operator $\mathcal{H}(\mathbf{k}, t)$:

$$\left(H(\mathbf{k}, t) - i\hbar \frac{\partial}{\partial t} \right) |u_{\alpha, \mathbf{k}}(t)\rangle = \varepsilon_{\alpha, \mathbf{k}} |u_{\alpha, \mathbf{k}}(t)\rangle, \quad (\text{B.13})$$

where $H(\mathbf{k}, t) = e^{-i\mathbf{k}\cdot\mathbf{r}} H(t) e^{i\mathbf{k}\cdot\mathbf{r}}$. Expanding the periodic state and Hamiltonian in Fourier series, $|u_{\alpha,\mathbf{k}}(t)\rangle = \sum_m e^{im\omega t} |u_{\alpha,\mathbf{k}}^{(m)}\rangle$ and $H(\mathbf{k}, t) = \sum_l H_l(\mathbf{k}) e^{il\omega t}$, and projecting onto the basis $e^{in\omega t}$, we obtain the infinite-dimensional matrix eigenvalue equation:

$$\sum_{m=-\infty}^{\infty} [H_{n-m}(\mathbf{k}) + n\hbar\omega\delta_{n,m}\mathbb{I}] |u_{\alpha,\mathbf{k}}^{(m)}\rangle = \varepsilon_{\alpha,\mathbf{k}} |u_{\alpha,\mathbf{k}}^{(n)}\rangle. \quad (\text{B.14})$$

The matrix elements $\mathcal{H}_{nm}(\mathbf{k}) = H_{n-m}(\mathbf{k}) + n\hbar\omega\delta_{n,m}\mathbb{I}$ define the Sambe space. The diagonal terms represent the time-averaged Hamiltonian shifted by photon energies (the "Stark ladder"), while off-diagonal terms H_{n-m} couple different Floquet replicas. In the high-frequency limit ($\hbar\omega \gg \|H_l\|$), off-diagonal couplings are perturbative, effectively decoupling the Floquet bands.

B.5 Derivation of the Floquet Berry Connection

The topological properties of a periodically driven system are encoded in the geometry of its Floquet modes. Following Furman [29], we consider the adiabatic evolution of the system with respect to the crystal momentum $\mathbf{k}$.

In the Sambe space, the inner product of two periodic states is defined by time-averaging over one period T :

$$\langle\langle u|v\rangle\rangle = \frac{1}{T} \int_0^T dt \langle u(t)|v(t)\rangle = \sum_m \langle u^{(m)}|v^{(m)}\rangle. \quad (\text{B.15})$$

For a non-degenerate isolated Floquet band n , the geometric phase accumulated under a slow variation of $\mathbf{k}$ is governed by the gauge potential in this extended space. Applying the gradient operator $\nabla_{\mathbf{k}}$ to the Fourier-expanded state $|u_n(t)\rangle = \sum_m e^{im\omega t} |u_n^{(m)}\rangle$, the cross-terms between different Fourier modes vanish upon time-averaging due to orthogonality. This reduces the extended-space Berry connection directly to the time-averaged integral:

$$\mathbf{A}_n(\mathbf{k}) = i\langle\langle u_n(\mathbf{k})|\nabla_{\mathbf{k}}|u_n(\mathbf{k})\rangle\rangle = \frac{i}{T} \int_0^T dt \langle u_n(\mathbf{k}, t)|\nabla_{\mathbf{k}}|u_n(\mathbf{k}, t)\rangle. \quad (\text{B.16})$$

This rigorously establishes that the Floquet topological invariants can be computed directly from the time-averaged geometry of the Floquet modes.

Appendix C: Derivations for Nonlinear Optics Theory

C.1 Classical Anharmonic Oscillator and Miller's Rule

The microscopic origin of optical nonlinearities can be modeled classically by an electron of mass m and charge $-e$ in an anharmonic potential. Including the lowest-order asymmetry (cubic term), the equation of motion under a driving field $E(t) = E_0 \cos(\omega t)$ and damping γ is:

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x + C_3 x^2 = -\frac{e}{m} E(t). \quad (\text{C.1})$$

We solve this perturbatively via $x(t) = x^{(1)}(t) + x^{(2)}(t) + \dots$. Using complex notation $E(t) = \frac{E_0}{2}(e^{-i\omega t} + \text{c.c.})$ and assuming $x(t) = \frac{1}{2}(X_1 e^{-i\omega t} + X_2 e^{-2i\omega t} + \text{c.c.})$, the first-order (linear) response is:

$$X_1 = \frac{-eE_0}{m[(\omega_0^2 - \omega^2) - i\gamma\omega]}, \quad \text{yielding} \quad \chi^{(1)}(\omega) = \frac{Ne^2}{\varepsilon_0 m[(\omega_0^2 - \omega^2) - i\gamma\omega]}. \quad (\text{C.2})$$

Substituting $x^{(1)}(t)$ into the anharmonic term $C_3 x^2$ generates a driving force at 2ω . Collecting the $e^{-2i\omega t}$ terms yields the second-order amplitude:

$$X_2 = \frac{-C_3 e^2 E_0^2}{2m^2 (\omega_0^2 - 4\omega^2 - 2i\gamma\omega) (\omega_0^2 - \omega^2 - i\gamma\omega)^2}. \quad (\text{C.3})$$

Equating the microscopic polarization $P^{(2\omega)} = -\frac{Ne}{2} X_2$ to the macroscopic definition $P^{(2\omega)} = \varepsilon_0 \chi^{(2)} \left(\frac{E_0}{2}\right)^2$, and substituting the linear susceptibilities from Eq. (C.2), we obtain **Miller's Rule**:

$$\boxed{\chi^{(2)}(2\omega; \omega, \omega) = \frac{mC_3\varepsilon_0^2}{N^2 e^3} [\chi^{(1)}(\omega)]^2 \chi^{(1)}(2\omega)} \quad (\text{C.4})$$

This demonstrates that $\chi^{(2)}$ is strictly proportional to the potential asymmetry C_3 (vanishing in centrosymmetric media) and exhibits strong resonance enhancement near ω_0 .

C.2 Symmetry and Selection Rules in High Harmonic Generation

In Section 3.5.1, we stated that centrosymmetric systems driven by a linearly polarized field emit only odd harmonics. Here, we provide a simple and rigorous derivation of this selection rule based on generalized parity symmetry.

Consider a system with a symmetric potential $V(-x) = V(x)$ interacting with a linearly polarized laser field. The total Hamiltonian is:

$$\hat{H}(t) = \frac{\hat{p}^2}{2m} + V(\hat{x}) + e\hat{x}F_0 \cos(\omega t). \quad (\text{C.5})$$

This Hamiltonian is periodic in time with period $T = 2\pi/\omega$. According to Floquet's theorem, the wavefunction can be written as:

$$\Psi_n(x, t) = e^{-i\varepsilon_n t/\hbar} \psi_n(x, t), \quad (\text{C.6})$$

where $\psi_n(x, t)$ is the periodic Floquet function, satisfying $\psi_n(x, t + T) = \psi_n(x, t)$.

For a symmetric potential, the Hamiltonian is invariant under a **generalized parity transformation** $\hat{\mathcal{P}}$ that combines spatial inversion ($x \rightarrow -x$) and a time shift by half a period ($t \rightarrow t + T/2$). Therefore, the Floquet states transform as:

$$\hat{\mathcal{P}}\psi_n(x, t) = \psi_n(-x, t + T/2) = \pm\psi_n(x, t). \quad (\text{C.7})$$

The macroscopic polarization or time-dependent dipole moment is given by $d(t) = \langle \psi_n(t) | \hat{x} | \psi_n(t) \rangle$. We want to find the m -th harmonic component d_m of this dipole moment, which is obtained via Fourier transform over one period:

$$d_m = \frac{1}{T} \int_0^T d(t) e^{im\omega t} dt. \quad (\text{C.8})$$

We can split this integral into two halves:

$$d_m = \frac{1}{T} \int_0^{T/2} [d(t) + d(t + T/2) e^{im\omega T/2}] e^{im\omega t} dt. \quad (\text{C.9})$$

Since $e^{im\omega T/2} = e^{im\pi} = (-1)^m$, this becomes:

$$d_m = \frac{1}{T} \int_0^{T/2} [d(t) + (-1)^m d(t + T/2)] e^{im\omega t} dt. \quad (\text{C.10})$$

Now, let us evaluate $d(t + T/2)$ using the generalized parity property (Eq. C.7):

$$\begin{aligned} d(t + T/2) &= \int dx \psi_n^*(x, t + T/2) x \psi_n(x, t + T/2) \\ &= \int dx \psi_n^*(-x, t) x \psi_n(-x, t) \\ &= \int dx' \psi_n^*(x', t) (-x') \psi_n(x', t) \\ &= -d(t). \end{aligned} \quad (\text{C.11})$$

Substituting $d(t + T/2) = -d(t)$ back into Eq. (C.10), we get:

$$d_m = \frac{1}{T} \int_0^{T/2} d(t) [1 - (-1)^m] e^{im\omega t} dt. \quad (\text{C.12})$$

From this expression, we can immediately see the selection rule:

- If m is **even**, $(-1)^m = 1$, so $1 - (-1)^m = 0$, which means $d_m = 0$. Even harmonics vanish.
- If m is **odd**, $(-1)^m = -1$, so $1 - (-1)^m = 2$, and the harmonic component d_m is generally non-zero.

Thus, for any centrosymmetric medium driven by a linearly polarized field, the generalized parity symmetry strictly forbids the emission of even-order harmonics, leaving only the odd harmonics in the spectrum. As discussed in Section 3.5.1, this selection rule can only be broken if the spatial inversion symmetry of the medium or the half-cycle symmetry of the driving field is dynamically broken.